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Emerging Financial Risks: Measurement and Market Relevance

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Chapter 1

Introduction

1.1 Background and Motivation

Financial risks develop through events that affect firms and the financial environment in which they operate and information about the risk reaches investors gradually, often before its economic consequences can be fully assessed.Articles document incidents as they occur and document their developments. As new information becomes available, the volume and content of reporting change.

These changes can be examined using quantitative methods for example: the number of articles associated with a risk provides a measure of the attention it receives, the language used in those articles offers further information about how the risk is being discussed.

This thesis studies five categories of emerging risk: Cyber Risk, Climate Risk, Pandemic Risk, Supply Chain Risk, and AI-Related Risk. The analysis uses newspaper articles published between 2014 and 2023 to build indicators that describe the evolution of reporting associated with each category.

The study begins with the measurement of these risks across the news corpus. Then it examines the behavior of the resulting indicators over time. The final part focuses on Cyber Risk and investigates whether changes in cyber-related news contain information about subsequent equity returns.

The financial application is inspired on the work of Cherubini et al. (2024), who examine the relationship between a text-based measure of cyber-risk attention and stock market returns. Their approach provides a starting point for investigating the market relevance of the indicators developed in this thesis.

1.2 Research Problem

Emerging risks are difficult to measure consistently because the information available about them changes as events unfold.

The initial problem concerns the identification of relevant news. The newspaper database contains articles covering many subjects, and the same risk can be described using different expressions.

A classification procedure is needed to identify reporting associated with each of the five categories.

The next problem concerns the measurement of changes in reporting. Consider two periods in which the number of cyber-related articles increases. In the first, newspapers may repeatedly cover an incident that has already been described. In the second, the increase may involve reports of previously unknown attacks affecting several firms. Both periods produce higher article counts, although the content of the incoming information differs. This motivates the construction of indicators that track the distinctiveness of new articles and changes in the language associated with each risk category.

The temporal dimension introduces a further question. A risk may receive a relatively high share of coverage for several months, while another may experience a sudden increase after a period of limited attention. Identifying these episodes requires comparing current reporting with the previous history of the same theme.

The thesis addresses these measurement problems through four indicators: Coverage, Novelty, Semantic Drift and Burst Detection.

The final research problem concerns the financial information contained in these measures. Once the indicators have been constructed, their relationship with market outcomes can be investigated. The empirical application focuses on Cyber Risk and examines whether changes in news-based indicators are associated with subsequent weekly equity portfolio excess returns.

1.3 Research Objectives

The first objective is to make sure to work with a news database suitable for the measurement of emerging risks. The material consists of articles published between 2014 and 2023 by five newspapers: the Financial Times, Wall Street Journal, New York Times, Los Angeles Times, and USA Today. After validation and deduplication, the database contains 2,249,255 articles.

The second objective is to identify reporting associated with the five selected risk categories. The classification procedure combines lexical rules with supervised text classification. Manually reviewed observations are used to evaluate the ability of the models to identify relevant articles. The resulting thematic scores provide the basis for the subsequent construction of the indicators.

The third objective is to represent the semantic content of the articles numerically. Document embeddings allow articles to be compared according to their textual content. These representations are used to examine how incoming news differs from earlier reporting and how the content associated with a risk category changes over time.

The fourth objective is to construct four measures of emerging risk from classified articles and their semantic representations. Here four are constructed: Coverage, Novelty, Semantic Drift, Burst Detection.
The indicators are constructed across the five risk categories. Their empirical analysis examines how

the measures evolve during 2015–2023.

The fifth objective is to investigate Cyber Risk in details to assess the market relevance of the cyber-risk indicators. Following the approach used by Cherubini et al. (2024), the thesis estimates models linking changes in cyber-related news measures to weekly equity excess returns.

The final analysis considers two equally weighted portfolios: the Diversified portfolio contains 195 firms from several industries and the Attacked + Security portfolio contains 76 firms, including 66 companies classified as having experienced cyberattacks and 10 cybersecurity firms.
The empirical models examine the relationship between lagged changes in the indicators and subsequent returns. An out-of-sample forecasting exercise then evaluates whether Coverage improves one-week-ahead return predictions relative to an autoregressive benchmark. Additional tests investigate the estimated relationships during periods of unusually intense cyber-risk reporting.

1.4 Contributions

The first contribution of the thesis is a common measurement framework for five emerging risk categories. The framework combines information about the frequency of reporting with information extracted from article content. Novelty describes how different incoming articles are from earlier news. Semantic Drift follows changes in the representation of each theme. Coverage measures the share of attention received by a risk, while Burst Detection identifies episodes of unusually strong increases in that attention. The empirical analysis shows how these measures behave across the five categories. Their time series reveal differences in the timing of changes in coverage and semantic content. The indicators also allow periods of intense reporting to be examined in relation to the information published during those periods.

The second contribution concerns the analysis of Cyber Risk. The thesis analizes the evolution of its indicators and identifies episodes in which changes in attention coincide with changes in semantic content. The analysis also documents periods in which the indicators follow different patterns.

The third contribution is the investigation of the financial relevance of cyber-risk news indicators. Building on the approach of Cherubini et al., the thesis examines the relationship between the constructed indicators and weekly equity returns for two portfolios with different compositions.

The results indicate that Coverage has the clearest statistical association with excess returns. Its six lagged changes are significant in both portfolios, including combinations with Novelty and/or Semantic Drift.
Adding Coverage increases adjusted R^2 by 3.61 percentage points for the Diversified portfolio and 4.45 percentage points for the Attacked + Security portfolio. The out-of-sample exercise also produces lower one-week-ahead root mean squared errors, with reductions of 0.85% and 1.20%, respectively, relative to the autoregressive benchmark.
The analysis of Burst episodes examines whether unusually intense cyber-risk attention changes the estimated relationship between news indicators and returns. The tests do not identify a statistically significant difference during these periods. The joint significance of Coverage remains when contem-

poraneous Burst weeks are excluded from the return regressions.

The financial results document a predictive association between cyber-risk Coverage and weekly excess returns within the portfolios and sample periods examined.

1.5 Structure of the Thesis

Chapter 2 reviews the literature studied before writing this thesis and the theoretical methods used in the financial analysis.
Chapter 3 describes the construction of the newspaper database.
Chapter 4 develops the thematic classification framework used to identify articles associated with the five risk categories. It also reports the evaluation of the classification procedure.
Chapter 5 presents the construction of document embeddings and explains how semantic representations are used to compare articles.
Chapter 6 develops the four emerging-risk indicators.
Chapter 7 examines the behavior of these indicators across the five risk categories during the sample period. It then focuses more closely on Cyber Risk, considering the evolution of its measures and the episodes identified in the news.
Chapter 8 investigates the market relevance of cyber-risk news. It estimates the dynamic relationship between the indicators and weekly equity excess returns, evaluates the forecasting contribution of Coverage, and examines the results during Burst episodes.
Chapter 9 contains the conclusions of the thesis.

Chapter 2

Literature Review and Theoretical Foundations

This thesis uses information from financial news to measure five risk themes and examine the relationship between cyber-risk news and weekly equity returns. The methods connect language representations to financial prices and use statistical tools for text classification and time-series inference. This chapter develops the theoretical foundations for Chapters 4–8.

2.1 Financial News in the Literature

2.1.1 Language representations and expected returns

Chen, Kelly, and Xiu study whether embeddings extracted from financial news by large language models help predict cross-sectional stock returns Chen et al., 2025. Their analysis examines the return-predictive information contained in language-model representations of financial news. The study serves as a benchmark on using news text as a numerical signal source in empirical asset pricing. An embedding is a numerical representation of a text where the coordinates are learned from language data. This representation makes it possible for a predictive model to learn patterns from all over the article, including relationships between phrases which have little in common in terms of vocabulary. Chen, Kelly, and Xiu leverage this property to establish the link between the content of news articles and their expected returns. This thesis uses document embeddings to compare an article with previous articles and observe the evolution of the content related to a risk theme. The embeddings enter the construction of Novelty and Semantic Drift in Chapter 6.

Didisheim, Kelly, Pourmohammadi, and Tian examine how predictable news content affects estimates of market responses to information Didisheim et al., 2026. They predict news embeddings using prevailing stock characteristics and study the residual component. In simplified notation, their distinction can be expressed as

$$\mathbf{e}_{i,t} = \widehat{\mathbf{e}}_{i,t} + \mathbf{u}_{i,t}, \tag{2.1}$$

where $\widehat{\mathbf{e}}_{i,t}$ is the predicted representation and $\mathbf{u}_{i,t}$ contains the part left unexplained by that prediction. The authors refer to this residual information as a news shock. Their results examine how the component of news that is unexpected given observable stock characteristics relates to subsequent returns.

Their analysis gives a precise interpretation to an informational surprise: it depends on what could have been predicted from the available conditioning variables. Chapter 6 measures semantic distinctiveness by comparing each article with earlier reporting. The historical comparison defines information relative to earlier articles. Didisheim et al. instead construct residual shocks relative to a prediction based on stock characteristics.

2.1.2 Cyber-risk news and equity returns

Cherubini, Della Lunga, Figà Talamanca, and Guardabascio construct a text-based index of cyber-risk attention from newspaper coverage and investigate its dynamic relationship with US sector-level equity returns Cherubini et al., 2024. Their analysis uses weekly observations and a vector autoregressive framework. Their empirical framework estimates the dynamic relationship between changes in cyber-risk news and equity returns across sectors.
Their paper motivates the market-relevance analysis in Chapter 8. Both studies use newspaper-based measures and weekly financial data, with six lags in the dynamic specification. The thesis examines Coverage alongside Novelty and Semantic Drift and tests their association with two equally weighted portfolios. Its return equation uses excess returns in levels and lagged changes in the news indicators. The lag coefficients measure conditional associations between past news and current portfolio returns.

2.2 Statistical Foundations of Text Classification

2.2.1 Lexical rules and weak supervision

The classification problem in Chapter 4 assigns thematic relevance to articles for five categories: cyber risk, climate risk, pandemic risk, supply-chain risk, and AI-related risk. A keyword baseline searches for terms associated with each theme. The terms selected for each rule determine which articles are captured. An article may describe a cyber incident without using a selected keyword; the same keyword may also appear in an unrelated discussion.
Weak supervision uses labeling rules to generate training information from text. A labeling function $\lambda_j(x_i)$ assigns evidence about article i from a specified textual pattern or contextual condition. Different functions may cover different parts of the text and can disagree on the same article. The resulting labels require validation against manually assessed observations.

2.2.2 TF–IDF and logistic regression

Term Frequency–Inverse Document Frequency (TF–IDF) maps documents to sparse vectors. A term receives greater weight when it appears frequently in a document and less weight when it is common throughout the corpus. A conventional expression is

$$\text{tfidf}(w, d) = \text{tf}(w, d)\ \text{idf}(w), \tag{2.2}$$

where w denotes a term and d a document. The thesis uses unigram and bigram features in its supervised thematic classifier. These features represent word occurrences and adjacent word pairs, allowing the model to associate expressions with particular risk categories.

Logistic regression relates a feature vector $\mathbf{x}_i$ to a binary label through the logistic link:

$$\Pr(Y_i = 1 \mid \mathbf{x}_i) = \frac{1}{1 + \exp[-(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta})]}. \tag{2.3}$$

The fitted coefficients determine how textual features contribute to the classifier score. With many sparse predictors, L_2 regularization penalizes large coefficients and limits sensitivity to the training sample. Class weighting changes the contribution of positive and negative examples to the fitting objective when their frequencies differ. Chapter 4 applies both choices.

Classification performance is assessed from true positives (TP), false positives (FP), and false negatives (FN). Precision measures the share of predicted positives that are correct, while recall measures the share of actual positives recovered. Their harmonic mean is the F_1 score:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2\ \text{Precision}\ \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{2.4}$$

Precision falls when the classifier assigns a risk label to unrelated articles; recall falls when it misses relevant articles. The supervised model also produces continuous thematic scores. Their interpretation depends on model training and calibration, so Chapter 4 treats them as relevance scores rather than calibrated probabilities of true theme membership.

2.3 Semantic Representations of Financial News

2.3.1 Document embeddings

Document embeddings transform an article into a vector $\mathbf{e}_i \in \mathbb{R}^d$ to produce contextual representations by processing the words in a document together. Their vector geometry makes comparisons between articles that describe similar events using different vocabulary possible. Chapter 5 uses the model `all-MiniLM-L6-v2`, which produces 384-dimensional document representations.

The embedding space provides a common numerical representation for comparisons across articles and publication dates. An article describing a ransomware attack, for example, may be close to another account of a network intrusion even if the two articles use different terms. This property makes embeddings suitable for studying how the content of risk-related reporting evolves.

2.3.2 Cosine similarity and temporal comparison

Cosine similarity measures the angle between two embedding vectors:

$$\text{sim}(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i^\top \mathbf{e}_j}{\|\mathbf{e}_i\|\,\|\mathbf{e}_j\|}. \tag{2.5}$$

For vectors normalized to unit length, cosine similarity equals their inner product. Values closer to one indicate greater directional similarity. The resulting score compares vector directions in the embedding space.

Temporal comparisons require a defined historical information set. A document published in week t can be compared with articles available before that week; a monthly thematic representation can likewise be compared with a historical reference. These comparisons provide the geometric basis for

the semantic measures developed in Chapter 6. The definitions of Novelty and Semantic Drift, together with the construction of Coverage and Burst Detection, are given there.

2.4 Econometric Models and Statistical Inference

2.4.1 Autoregressive models and VAR specifications

The financial analysis studies weekly portfolio excess returns $R_{P,t}$ and the weekly changes in news indicators $\Delta I_t = I_t - I_{t-1}$. An autoregressive model uses lagged values of a series to account for its own time dependence:

$$R_{P,t} = a_P + \sum_{\ell=1}^{p} b_{P,\ell} R_{P,t-\ell} + \varepsilon_{P,t}. \tag{2.6}$$

An AR(6) model serves as the forecasting benchmark in Chapter 8. The augmented specification adds six lags of the change in cyber-risk Coverage.
A vector autoregression (VAR) models several series jointly. For a vector $\mathbf{Y}_t$, a VAR(p) is

$$\mathbf{Y}_t = \mathbf{a} + \sum_{\ell=1}^{p} A_\ell \mathbf{Y}_{t-\ell} + \mathbf{u}_t. \tag{2.7}$$

The thesis estimates bivariate VAR(6) specifications with $\mathbf{Y}_{P,t} = (R_{P,t}, \Delta I_t)^\top$. The return equation is

$$R_{P,t} = a_P + \sum_{\ell=1}^{6} b_{P,\ell} R_{P,t-\ell} + \sum_{\ell=1}^{6} c_{P,\ell} \Delta I_{t-\ell} + \varepsilon_{P,t}, \tag{2.8}$$

where the coefficient $c_{P,\ell}$ measures the association between a change in the indicator ℓ weeks earlier and the current excess return, conditional on the other regressors. Models containing several indicators include a separate set of six lags for each measure. The equations are estimated by ordinary least squares (OLS), which minimizes the sum of squared residuals. For a regression $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, the OLS estimator is $\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$ when the inverse exists.

2.4.2 Wald tests and HAC covariance estimation

A Wald test evaluates linear restrictions on estimated coefficients. In general form, the null hypothesis is $H_0 : R\boldsymbol{\beta} = \mathbf{r}$. Given an estimated covariance matrix $\widehat{V}$, the test statistic is

$$W = (R\widehat{\boldsymbol{\beta}} - \mathbf{r})^\top (R\widehat{V}R^\top)^{-1} (R\widehat{\boldsymbol{\beta}} - \mathbf{r}). \tag{2.9}$$

Under standard regularity conditions, W converges under the null to a chi-squared distribution with degrees of freedom equal to the number of independent restrictions. In the return equation, the individual-indicator null is $H_0 : c_{P,1} = \cdots = c_{P,6} = 0$. Combined models allow the joint significance of several lag blocks to be tested. Conditional tests retain the other news indicators while restricting one block of coefficients.

Weekly regression errors may exhibit changing variance and serial dependence. Heteroskedasticity- and autocorrelation-consistent (HAC) covariance estimation accounts for these features when computing

coefficient uncertainty. A generic long-run covariance estimate combines contemporaneous covariance with weighted lagged autocovariances:

$$\widehat{\Omega} = \widehat{\Gamma}_0 + \sum_{\ell=1}^{L} w_\ell (\widehat{\Gamma}_\ell + \widehat{\Gamma}_\ell^\top). \tag{2.10}$$

The bandwidth L determines the number of included autocovariance lags. Chapter 8 uses HAC(6) for its main Wald tests and HAC(12) as a bandwidth sensitivity check. HAC changes the estimated coefficient covariance matrix and the resulting test statistics. The OLS point estimates remain unchanged. A small Wald-test p-value indicates that the data are inconsistent with the imposed restrictions under the chosen inferential framework.

2.5 Forecast Evaluation and Conditional Relationships

2.5.1 Expanding-window forecasts and nested-model comparison

An expanding-window exercise estimates the forecasting model using observations available up to each forecast origin. The next observation is predicted, the sample is extended, and the procedure is repeated. This design evaluates the information available to the model at the time each forecast is formed.
Chapter 8 compares one-week-ahead forecasts from an AR(6) benchmark with forecasts from an augmented model containing six lagged changes in Coverage. Let $e_{0,t}$ and $e_{1,t}$ be the corresponding forecast errors. Root mean squared error (RMSE) and out-of-sample R^2 are

$$\mathrm{RMSE}_j = \sqrt{\frac{1}{T}\sum_{t=1}^{T} e_{j,t}^2}, \qquad R^2_{\mathrm{OOS}} = 1 - \frac{\sum_t e_{1,t}^2}{\sum_t e_{0,t}^2}. \tag{2.11}$$

A positive out-of-sample R^2 indicates lower squared forecast error for the augmented model over the evaluation period.
The Clark–West test addresses comparisons between nested forecasting models. Estimating additional coefficients can increase the larger model's squared forecast errors under the null of no incremental predictive content. The test adjusts the loss differential for the difference between the two forecasts. If $\widehat{R}_{0,t}$ and $\widehat{R}_{1,t}$ denote the benchmark and augmented forecasts, the adjusted differential is

$$f_t = e_{0,t}^2 - e_{1,t}^2 + (\widehat{R}_{0,t} - \widehat{R}_{1,t})^2. \tag{2.12}$$

The one-tailed test on its mean assesses the improvement in forecasting accuracy made by the enhanced model after the nesting correction. The thesis uses HAC(6) tests on this difference. This test allows an inferential assessment of the forecasting variable in addition to the reported RMSEs.

2.5.2 Burst interactions and sensitivity checks

Interaction regressions check whether a fitted relationship depends on the state. Let $S_{I,t}$ represent lagged changes in a news indicator, and let B_{t-1} represent a Burst state at time *t-1*. A general specification is

$$R_{P,t} = \alpha_P + \beta_P S_{I,t} + \gamma_P B_{t-1} + \delta_P S_{I,t} B_{t-1} + \mathbf{z}_t^\top \boldsymbol{\theta}_P + \varepsilon_{P,t}, \tag{2.13}$$

where $\mathbf{z}_t$ collects the other controls. The slope on $S_{I,t}$ is β_P in ordinary weeks and $\beta_P + \delta_P$ following a Burst state. Testing $H_0 : \delta_P = 0$ assesses whether the two slopes differ. Chapter 8 uses HAC(6) inference for these regressions and examines the sensitivity of its Coverage results to the exclusion of contemporaneous Burst weeks.

Chapter 8 applies these tests to weekly portfolio returns and reports the out-of-sample comparison between the AR(6) benchmark and the model augmented with lagged Coverage.

Chapter 3

Database Construction and Data Preprocessing

The news database serves as the basis for the empirical analysis of the present thesis. The database includes articles which were written between 2014 and 2023 and consist of material from five different newspapers: Financial Times (FTXX), Wall Street Journal (WSJX), New York Times (NYTX), Los Angeles Times (LATX) and USA Today (USAT).

Raw files are first placed in a staging table in order to allow for the checking of the content of the files before they are added to the database. The checking concerns basic details, including the publication date, newspaper, language, title, and article text. These records are then put into a standard format.

Duplicate texts require a special approach. Duplicates can appear multiple times within the original database, but are not always identical. For this reason, the database allows for the distinction of ex-act duplicates and near-duplicates through two separate approaches.

These choices matter for the empirical analysis. If the same article appears several times, for example, it receives more weight than it should when news coverage is measured. Errors in dates or source information can create similar problems when articles are compared across newspapers or over time. The cleaned corpus used in the following chapters is constructed after these issues have been addressed.

3.1 Dataset and Database Architecture

3.1.1 News Corpus

The unit of observation is an individual news article. The source files contain both textual information and article-level metadata: the article title and body, publication date, language, source, an original sentiment variable, a duplicate indicator, a data-type field, and the article URL.
The staging table contains 2,374,643 records. After validation and transfer to the main table, `news_articles` contains 2,373,999 records. Deduplication is performed subsequently on this set of articles.

3.1.2 Database Architecture

The news corpus is stored in PostgreSQL, in the `research` database. A dedicated schema, `risk_emergence`, contains the tables and views developed for the project. Its structure keeps the original ingestion layer separate from the data and variables produced later in the analysis.

There are two tables that play a direct role in the initial creation of the news corpus. The first one is `news_articles _staging`, where the records from the source files are temporarily stored. The second one is `news_articles`, which is the core article table used by the rest of the database.

The structure of the staging table is intentionally very simple. All the fields are stored as text and there are no constraints on them at this stage. Therefore, records can be inserted into the table without checking dates, sources and other fields and converting them to their proper types. Additionally, the table contains `_raw_line_number` and `_source_file`. These fields refer to the original row and file of each record in the source files. This allows tracking down problematic observations to their source data.

After all of the necessary transformations and calculations are done, the data goes into `news_articles`. This table keeps the original article data and some additional columns generated in the process of data preprocessing. Some of them include `word_count`, `char_count`, `language`, `is_duplicate`, and `content_hash`. The last one is calculated on the basis of the article text and will be useful in detecting exact duplicates. In addition, `news_articles` stores the full text of the title and article body in `title_tsv` and `body_tsv`.

However, the database does not end at these two tables. There are other objects storing data about newspapers, theme scores, article embeddings, mentioned firms, and aggregated metrics. All of them are related to `news_articles`, which serves as the core table for the news part of the project.

3.2 Data Ingestion and Validation

3.2.1 Raw Data Ingestion

The original news data are stored in UTF-8 encoded CSV files using the vertical bar (`|`) as delimiter. The ingestion procedure is implemented in Python and proceeds through the staging table before records are transferred to the final database.

Before importing the observations, the expected CSV header is checked against the nine fields: Title, Body, Date, Language, Source, Sentiment, IsDuplicate, dataType, Url. Each raw observation is required to contain exactly nine fields. Rows that do not satisfy this structural requirement are not transferred as valid observations and are recorded as malformed rows.

The ingestion is performed in batches rather than by loading the entire corpus into memory. During this process, the original row number and source file are appended to each record through the variables `_raw_line_number` and `_source_file`. The resulting observations are inserted into

`risk_emergence.news_articles_staging`.
This staging-first design separates data acquisition from data transformation. In particular, observations can be inspected and validated before type conversions and normalisation rules are applied.

3.2.2 Staging Validation

Once the raw observations have been imported, the staging table is subjected to a series of validation checks. Given the size of the corpus, the validation is performed incrementally using a server-side cursor rather than loading all observations simultaneously into memory.
The validation procedure examines the main variables and it verifies that:

- publication dates can be parsed into an accepted date format;
- the newspaper source can be mapped to a recognised source;
- the original sentiment value is either missing or numerically interpretable;
- the original duplicate indicator can be interpreted as a Boolean value;
- the article title is not empty;
- the article body is not empty;
- the language value can be classified consistently.

Multiple date formats are accepted during parsing in order to accommodate differences in the raw source files. The validation stage therefore separates the question of whether a date can be interpreted from the subsequent normalisation of that date into a common database representation.
Similarly, the sentiment variable is validated only as an original metadata field. No new sentiment analysis is performed during this stage. Missing sentiment values are allowed, while non-missing observations must be convertible to a numerical representation.
The original duplicate indicator is also checked independently from the deduplication procedure described later in this chapter. This distinction is important because the deduplication performed for the thesis does not rely exclusively on the duplicate information supplied in the raw files.
After transfer to the final database, quality-control queries confirmed the presence of 2,373,999 observations and no missing values in the key fields `title`, `body`, `publication_date`, and source information. The observed date range corresponds to the intended sample period, from January 1, 2014 to December 31, 2023.

3.3 Data Normalisation

Once validated, the records are then reformatted for ingestion into the final database. Data normalization involves date parsing, source mapping, language normalization, data type conversion and creation of variables necessary for future analysis.

For example: the original source codes are matched with the corresponding newspaper id codes within the database; language data is converted to a consistent format which will then be used to isolate

English-language observations; finally, the dates are converted to database date format.

The final table also includes variables that measure the textual content of the articles. Such variables facilitate data quality inspection as well as future text processing. One such variable, `content_hash`, is computed for each article from the text body through the SHA-256 hashing function. For an article with text B_i, the hash will be

$$h_i = \text{SHA256}(B_i). \tag{3.1}$$

The content hash provides a deterministic representation of the article body and is the basis for the exact-deduplication procedure described in Section 3.4.

3.4 Deduplication

Duplicate news articles are another possible cause of measurement error. Multiple copies of an article would artificially inflate the counts of particular themes, thus impacting the metrics of coverage, novelty, and temporal variation. For this reason, deduplication is carried out prior to the use of the corpus in the following NLP and embedding processes.
One measure of deduplication is not enough for this task. Identical articles can be determined using deterministic means, but articles that are different due to slight editorial adjustments are to be identified through the process of comparing similarities. The deduplication process includes two steps:

$$\text{Exact deduplication} \quad \longrightarrow \quad \text{Near-duplicate detection.} \tag{3.2}$$

3.4.1 Exact Duplicate Detection

Exact duplicates are identified by the SHA-256 hash of the article body as in Equation 3.1. Two observations with the same content hash are considered as copies of the same textual content.
Only one article from each group of articles with the same `content_hash` is kept. The articles are ordered by `article_id`, and the observation with the smallest identifier is kept. All remaining observations in the group are flagged as exact duplicate.

Conceptually, if G_h denotes the set of articles sharing hash h, the retained article is

$$i_h^* = \arg\min_{i \in G_h} \texttt{article_id}_i. \tag{3.3}$$

All other observations belonging to G_h are flagged as exact duplicates.

Results This procedure identified 70,964 duplicate so, starting from 2,373,999 articles, the exact-deduplication stage therefore left 2,303,035 articles eligible for the subsequent near-duplicate analysis.
It is important to note that duplicate observations are not deleted from the database. The `is_duplicate_flag` records their status while preserving the original records. This allows the complete ingestion history to be retained and makes the deduplication procedure auditable.

3.4.2 Near-Duplicate Detection

Exact hashing cannot identify articles that contain the same content but differ slightly in the text. Examples include limited insertions or deletions, minor editorial changes, or alternative versions of essentially the same article. A second deduplication stage was therefore implemented using MinHash and Locality Sensitive Hashing (LSH).

Only the 2,303,035 observations not already classified as exact duplicates enter this stage.

Each article body is represented as a set of consecutive five-word shingles. If the tokenised body of an article is

$$(w_1, w_2, \ldots, w_n), \tag{3.4}$$

its shingle representation is

$$S_i = \left\{(w_j, w_{j+1}, w_{j+2}, w_{j+3}, w_{j+4})\right\}_{j=1}^{n-4}. \tag{3.5}$$

MinHash signatures are then computed using 128 permutations. The signatures provide an efficient approximation of the similarity between the shingle sets and are indexed using MinHash LSH with a threshold of 0.90.

LSH approximate output is not used directly to classify an observation as a duplicate. Instead, candidate pairs identified through LSH are subjected to an exact Jaccard-similarity calculation.

For two articles represented by shingle sets A and B, Jaccard similarity is defined as

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}. \tag{3.6}$$

A candidate is classified as a near-duplicate only when

$$J(A, B) \geq 0.90. \tag{3.7}$$

The high similarity threshold makes the procedure deliberately conservative. Its purpose is to identify copies or near-copies of the same textual content rather than articles that merely discuss the same event or topic. In particular, independently written articles from different newspapers covering the same event should not be classified as duplicates solely because they are semantically related.

Results The LSH stage generated 59,903 candidate matches, of which 59,418 were subjected to exact Jaccard comparisons. The final procedure identified 53,780 additional near-duplicate articles satisfying the 0.90 Jaccard threshold.

3.4.3 Construction of the Final Analytical Corpus

After both deduplication stages, the number of articles available for the subsequent empirical analysis is

$$N_{\text{valid}} = N_{\text{total}} - N_{\text{exact}} - N_{\text{near}} \tag{3.8}$$
$$= 2,373,999 - 70,964 - 53,780 \tag{3.9}$$
$$= 2,249,255. \tag{3.10}$$

This corresponds to approximately 5.25% of the corpus.

Observations were not excluded in any way using the separate `is_excluded` flag at the end of this preprocessing step. The difference between the normalised database and the analytical corpus can thus be explained by only the two deduplication techniques mentioned above.
The resultant corpus consists of 2,249,255 articles and forms the document universe on which the following stages of the analysis rely. Practically, the observations are made available via the `risk_emergence.v_valid_articles` view, offering a unified input for the following stages of thematic scoring and document representation.
This approach allows preserving both the initial database and the process of the construction of the analytical sample. The exact duplicates, near-duplicates, and any other exclusions are encoded as separate fields of the database, rather than being deleted from the dataset.

Chapter 4

Thematic Relevance Scoring

The first empirical objective of this study is to transform the textual content of news articles into quantitative measures of their relevance to a predefined set of financial risk categories. The analysis focuses on five candidate themes: cybersecurity risk, climate and environmental risk, pandemic risk, supply-chain risk, and artificial-intelligence-related risk.
For each article (i), theme (k), and scoring methodology (m), a continuous relevance score is defined as

$$S_{i,k}^{m} \in [0, 1], \tag{4.1}$$

where higher values indicate stronger evidence that article (i) is related to risk category (k).

This distinction is important: each article is not necessarily assigned to a single topic as it may simultaneously contain information about multiple risk categories, and the intensity of its relevance may vary considerably. The thematic-scoring problem is therefore intrinsically multi-label and continuous.

Moreover, the resulting scores depends on the methodology used. A keyword-based score measures lexical evidence, a weakly supervised score measures the intensity of rule-based contextual evidence, and the supervised model produces a continuous score derived from the logistic classifier.
The use of multiple methodologies serves two purposes. First, it provides alternative measurements of the same latent construct, namely thematic relevance. Second, it allows the strengths and limitations of increasingly sophisticated text-analysis approaches to be assessed empirically.
The three final methodologies are:

1. keyword-based scoring;
2. weakly-supervised scoring;
3. supervised scoring.

The process of going from the first method to the third one is characterized by an incremental shift from lexical rules to context rules and ultimately from context rules to the statistical ones based on human labeling.

4.1 Thematic Framework

The five candidate risk categories were defined ex ante and stored in the project database together with an initial set of seed terms. The categories are:

1. Cybersecurity Risk;
2. Climate / Environmental Risk;
3. Pandemic Risk;
4. Supply Chain Risk;
5. AI-Related Risk.

Seed terms include expressions such as *cybersecurity*, *data breach*, and *ransomware* for cyber risk; *climate change*, *carbon*, and *emissions* for climate risk; *pandemic*, *COVID*, and *epidemic* for pandemic risk; *supply chain*, *shortage*, and *logistics* for supply-chain risk; and *artificial intelligence*, *machine learning*, *algorithmic*, and *generative AI* for AI-related risk.
These are the words that give us our starting point for the definition of the categories, however, they do not represent the thematic classification itself. Lexical presence does not always mean risk significance. The above-mentioned deficiency explains why we have to develop three score methods.

4.2 Keyword-Based Scoring

The keyword method's purpose is to quantify direct lexical evidence connecting an article to a candidate risk category.
If an article contains expressions associated with a theme, its score increases with the strength of the lexical match in the full text. If no thematic expression appears, the score is zero.
Therefore, the keyword method is treated as a baseline since lexical matching cannot reliably distinguish substantive risk discussion from incidental or ambiguous uses of the same expressions.
For example, an environmental term may occur in a context unrelated to financial climate risk, while an article may discuss a risk substantively without using the exact expressions contained in the initial seed list.

4.2.1 Keyword Score

The implementation depend on PostgreSQL full-text search. Single-word seed terms are converted using `plainto_tsquery`, whereas multi-word expressions are represented using `phraseto_tsquery`. The resulting expressions are combined through an OR query and matched against the article body.
For an article (i) and theme (k), the raw lexical relevance measure is obtained through PostgreSQL's `ts_rank`. Articles with no thematic match receive

$$S_{i,k}^{KW} = 0. \tag{4.2}$$

For matching articles, the raw rank is normalized using the 99th percentile of the rank distribution among matching observations for the corresponding theme. The final score is

$$S_{i,k}^{KW} = \min\left(\frac{R_{i,k}}{P99_k}, 1\right) \tag{4.3}$$

where $R_{i,k}$ is the raw `ts_rank` value and $P99_k$ is the 99th percentile of positive ranks for theme k. The use of the 99th percentile rather than the maximum makes the normalization less sensitive to extreme observations and values exceeding this reference point are clipped at one.

4.2.2 Limitations

Manual inspection revealed an important limitation of direct lexical matching: the same expression may have substantially different meanings depending on context.
The keyword score therefore measures *lexical thematic evidence*, not a complete semantic assessment of whether an article discusses the corresponding financial risk.
This motivated the introduction of contextual rules through weak supervision.

4.3 Weakly-Supervised Scoring

This method introduced labeling functions that evaluate different forms of textual evidence, including thematic expressions, explicit risk terminology, title evidence, local proximity, and contextual patterns. The key idea is that the relevance of an article to a given risk category depends also on how specific thematic terms are and on the context in which they appear.

The approach is called *weak supervision*. Thematic signals are generated automatically through predefined, imperfect rules and were developed and refined using human inspected observations during the development stage.

4.3.1 Contextual Labeling Functions

The labeling functions were built separately for each of the five risk categories. Each risk has its own vocabulary and, more importantly, its own sources of ambiguity.

In the case of cyber risk, there were specific terms which were strong signals for indicating the presence of a risk. Ransomware and data breach can serve as an example here. The same goes for the terms which indicate hacking, cyberattacks, security breaches, or cyber threats. Some terms, on the other hand, could be misleading when used independently; therefore, the rules also considered the title and context in which the term occurs. The appearance of a term associated with cyber risk next to words associated with attacks, vulnerabilities, threats, losses, or damages is a very good indicator of an article describing an actual risk.

Climate risk was more challenging because environmental language occurs in many articles that do not refer to climate risks. Terms such as "climate risk" and "climate crisis" are clear signals here. The rules also applied to physical and transition risks. Physical risk involved climate events that have been connected with some kind of damage or economic effects. Transition risk included topics like climate policy, regulation, and the energy transition. Generic environmental terms were more conservative.

These terms on their own did not constitute a signal. They had to occur in proximity to damage, disruption, exposure, risk or economic effects.

As far as pandemics and the associated risks go, direct mention of pandemics or infectious diseases were the starting point. It was important to consider what happened due to the occurrence of the health event. Restrictions, public health impact, economic consequences and disruption of some sort were considered as a part of the event. The importance of using these terms stemmed from the possibility of mentioning epidemic words as a part of something else entirely or as a figure of speech. Specific pandemic term was an indication on its own. Otherwise, the context of using the word should have shown that the article was about the health crisis and its consequences.

For supply-chain risk, there is a similar issue with commonly used business jargon. Such words as “shipping”, “suppliers”, “logistics”, “shortage” are used often without any underlying supply chain issues being discussed. On their own, those words were insufficient. However, specific expressions such as “supply chain disruption” were better indicators. The terms included shortages, logistical and transportation issues, supply problems, production bottlenecks, limitations, and disruptions of goods movement or availability. In this case, the context around the term played a major role. The mention of suppliers, for instance, would meant much more when it appeared next to language describing delays, shortages, or production problems.

More strict criteria had to be applied to articles on AI risk. Indeed, an article can discuss artificial intelligence at length without mentioning anything about AI risk. The article might simply describe some technology, a company, an AI system, or some new AI application. Therefore, the inclusion of AI terms alone was not sufficient. Phrases like "AI risk," "AI safety," and "algorithmic bias" were deemed very significant. General AI references had to be near some kind of language related to threats or adverse consequences, like misinformation, threats, privacy, cybersecurity, copyright, safety, bias, or job loss. Additionally, there was another issue with articles on AI. While it can be the source of some threat, it can also be used to prevent it. These two types of articles had to be differentiated somehow. If an article discusses AI detection of cyberattacks, prediction of climatic events, or handling supply chain issues, then it cannot be an article on AI risk. Instead, the rules had to reflect this difference rather than counting any article where AI and risk-related terms co-occurred.

The same basic idea was used for all five categories. Very specific terms could provide enough evidence on their own. Broader or more ambiguous terms needed support from the surrounding text. We introduced this distinction after looking at the mistakes made by earlier versions of the labeling system. Many of the false positives came from keywords that were technically related to a risk category but were being used in a different context.

The frozen Weak Supervision V3 rules were developed according to this logic. They were more sophisticated than mere lists of keywords. They considered how specific a term was, what appeared around it, and whether related concepts occurred close enough to suggest that they were actually connected in the article.

4.3.2 Iterative Development and Error Analysis

The weak-supervision system was developed iteratively. The general workflow was

$$\text{initial rules} \rightarrow \text{manual validation} \rightarrow \text{error analysis} \rightarrow \text{rule revision} \rightarrow \text{re-evaluation}. \tag{4.4}$$

The rules were refined step by step by looking at the false positives and false negatives produced by each version. When the same type of error appeared more than once, the rule behind it was changed rather than the individual article. The aim was to make each change useful for new articles as well, not only for the examples already examined during the error analysis. After several rounds of testing and adjustment, the final weak-supervision rules were frozen and used to build the continuous relevance score.

4.3.3 Weak-Supervision Score

The frozen Weak Supervision V3 rules provide the qualitative structure for the continuous weakly supervised score. The final continuous specification combines three quantities derived from the contextual evidence identified in an article:

- the frequency of strong thematic evidence;
- the density of strong thematic evidence;
- the proximity between relevant thematic and risk expressions.

Let the strong-term frequency be denoted by f, the strong-term density per 1,000 text units by d_s, and the minimum relevant proximity distance by d. Three normalized components are constructed:

$$F = 1 - e^{-f}, \tag{4.5}$$

so that additional occurrences increase the score while their marginal contribution progressively decreases,

$$D = \frac{d_s}{1 + d_s}, \tag{4.6}$$

which maps the density of strong evidence to the interval [0,1) and

$$P = \max\left(0, 1 - \frac{d}{50}\right) \tag{4.7}$$

Under this specification, closer thematic and risk expressions generate stronger proximity evidence, while pairs separated by 50 or more text units receive no contribution from the proximity component. The proximity window of 50 is part of the frozen Weak Supervision V3 configuration selected during model development.

The three components are combined differently across themes, consistently with the theme-specific structure of the contextual rules. For cybersecurity, climate, and supply-chain risk, the continuous score is

$$S_{i,k}^{WS} = 1 - (1 - F)(1 - P)(1 - 0.35D), \tag{4.8}$$

For pandemic risk, the frozen specification combines frequency and density without the proximity component:

$$S_{i,\text{Pandemic}}^{WS} = 1 - (1 - F)(1 - 0.35D). \tag{4.9}$$

For AI-related risk, the continuous score is based on the proximity component:

$$S_{i,\text{AI}}^{WS} = P. \tag{4.10}$$

This more restrictive specification is consistent with the particular ambiguity of AI terminology. Frequent references to artificial intelligence do not necessarily constitute evidence of AI-related risk; what matters is whether AI-related expressions occur in the relevant adverse-risk context.

The coefficient 0.35 represents the density-reinforcement parameter of the continuous specification selected during development. Together with the proximity window of 50, it was fixed as part of the final configuration before evaluation on the independent final test set.

The resulting weakly supervised score lies in the interval [0,1]. It should not be interpreted as a calibrated probability that an article belongs to a given risk category. Instead, it represents a continuous rule-based measure of the strength of thematic evidence identified by the frozen Weak Supervision V3 specification.

4.4 Supervised Scoring

The supervised approach differs conceptually from the previous two methods.
Keyword scoring relies on explicit lexical matching. Weak supervision relies on contextual rules specified during model development. The supervised classifier instead learns statistical regularities from manually labelled article–theme observations.
Each training observation is an article–theme pair *(i,k)* associated with a human label

$$y_{i,k} \in 0, 1. \tag{4.11}$$

The objective is to learn which textual patterns distinguish substantively relevant article–theme pairs from irrelevant ones.

4.4.1 Text Representation

The final supervised architecture uses a `FeatureUnion` combining three complementary text representations.

First is a shared word-level TF–IDF representation using word unigrams and bigrams, English stop-word removal, sublinear term frequency, a minimum document frequency of two, a maximum document-frequency threshold of 0.95, and a maximum of 25,000 features.

Second is a shared character-level TF–IDF representation using $char_wb$ character 3- to 5-grams, a minimum document frequency of two, sublinear term frequency, and a maximum of 25,000 features.

Third, explicit theme–text interactions are introduced using unigram and bigram features generated through a custom analyzer generated through a custom analyzer, with a minimum document frequency of two, a maximum document frequency of 0.95, a maximum of 50,000 features, sublinear term frequency, and L_2 normalization.

The theme–text component is important because the same article is evaluated with respect to multiple candidate risks. A textual expression may therefore provide strong evidence for one theme while being irrelevant to another.

4.4.2 Logistic Regression

The combined TF–IDF representation is passed to a logistic-regression classifier.
For a feature vector (x), the model computes

$$z = \beta_0 + \sum_{j=1}^{p} \beta_j x_j, \tag{4.12}$$

followed by the logistic transformation

$$p(y = 1|x) = \frac{1}{1 + e^{-z}}. \tag{4.13}$$

The frozen model uses:

- $C = 4$;
- L_2 regularization;
- the `liblinear` solver;
- balanced class weights;
- a maximum of 2,000 iterations;
- random seed 42.

Balanced class weights were used because positive thematic observations were substantially less frequent than negative observations.
The frozen classification threshold is

$$\tau = 0.34. \tag{4.14}$$

Thus, for binary evaluation,

$$\hat{y}_{i,k} = \mathbb{I}\left[S_{i,k}^{SUP} \geq 0.34\right]. \tag{4.15}$$

For the full corpus, the model keeps the continuous `predict_proba` score instead of using only the final 0/1 classification. This score shows how strongly the model associates an article with a given risk theme. Since no additional probability calibration is applied, it is used as a continuous model score rather than as an estimate of the article's true probability of belonging to that risk category.

4.5 Full-Corpus Production

The final analytical corpus contains

$$N = 2,249,255 \tag{4.16}$$

valid and non-duplicate articles.
Each article is evaluated against all five candidate themes. Therefore, each methodology produces

$$2,249,255 \times 5 = 11,246,275 \tag{4.17}$$

article–theme scores.
With three frozen methodologies, the complete final scoring view contains

$$11,246,275 \times 3 = 33,738,825 \tag{4.18}$$

method–article–theme observations.

These scores tell us *which risk category an article is associated with and to what extent*. The scores do not, however, represent the semantics of the article generally.
The next chapter discusses document embeddings, which embed each article into a continuous semantic space independently of the five predefined risk categories.

Chapter 5

Document Embeddings

5.1 Introduction and Motivation

The thematic relevance scores developed in Chapter 4 provide a representation of the relationship between each article and five predefined financial risk categories. They answer a specific question: *how strongly is article related to risk category?*
This representation is necessarily conditional on the categories defined ex ante. It cannot, by itself, capture the complete semantic content of an article or measure whether two articles discuss similar information using different terminology. A second representation is therefore required.

For each article (i), the objective of the present chapter is to construct a document-level semantic vector

$$E_i \in \mathbb{R}^d, \tag{5.1}$$

where d denotes the embedding dimension.
On the other hand, compared to thematic scores, the individual entries of E_i cannot be interpreted as anything in particular. The key is the relative positioning of the vectors in the embedded space: articles which are semantically close should be positioned near each other.
This is a fundamental point about the empirical procedure:

$$S_{i,k,m} \quad \rightarrow \quad \text{theme-specific relevance,} \tag{5.2}$$

$$E_i \quad \rightarrow \quad \text{general semantic representation.} \tag{5.3}$$

Document embeddings are subsequently used to construct measures of semantic novelty and semantic drift.

5.2 Sentence-Transformer Representation

The embedding model turns each text into a numeric vector of its semantics.
The general concept here is that articles with the same semantic meaning will be similar in their vectors regardless of whether they use the same wording. That means the same event can be described in

different ways in two articles, which are still describing the same situation.
In news data, such a problem can arise since the same piece of information can be covered in different ways. So, focusing solely on the common words in these articles would miss the semantic similarity between the articles.
Thus, the project design was based on two Sentence-Transformer approaches:

- `sentence-transformers/all-MiniLM-L6-v2`;
- `sentence-transformers/all-mpnet-base-v2`.

Rather than selecting one model a priori, both candidates were benchmarked on actual articles from the corpus.

5.3 Model Selection

5.3.1 MiniLM and MPNet

The two models differ in dimensionality, context capacity, and computational cost.
MiniLM produces

$$E_i \in \mathbb{R}^{384}, \tag{5.4}$$

with a maximum sequence length of 256 tokens.
MPNet produces

$$E_i \in \mathbb{R}^{768}, \tag{5.5}$$

with a maximum sequence length of 384 tokens.
The higher dimensionality of MPNet does not by itself imply a superior representation for the present application. Model selection must instead account for semantic representation, document length, computational cost, and the scale of the corpus.

5.3.2 Real-Corpus Benchmark

The two models were benchmarked using 1,000 real articles from the analytical corpus.
MiniLM achieved an encoding throughput of approximately 172.62 articles per second, whereas MPNet processed approximately 20.45 articles per second. MPNet was therefore approximately 8.4 times slower in the benchmark.
MPNet has a longer context window, but this only slightly reduced the number of articles that exceeded the limit. Approximately 94.4% of articles exceeded MiniLM's 256-token limit, while approximately 89.9% exceeded MPNet's 384-token limit.
Thus, moving from MiniLM to MPNet did not solve the fundamental long-document problem. At the same time, MPNet substantially increased computational cost and doubled the dimensionality of the stored vectors.

MiniLM was therefore selected as the embedding backbone.

This decision should not be interpreted as a general claim that MiniLM is superior to MPNet. It is a corpus-specific engineering and methodological choice based on the trade-off observed in the present application.

5.4 The Long-Document Problem

The news articles in the corpus are substantially longer than the native MiniLM context window.
In the initial document-length benchmark the mean article length was approximately 930 tokens and the 95th percentile exceeded 2,000 tokens.
Consequently, the standard pipeline

$$\text{title + body} \rightarrow \text{MiniLM} \rightarrow \text{automatic truncation} \tag{5.6}$$

would discard a substantial portion of the text for the overwhelming majority of articles.
This would be problematic for the present application. The objective is to obtain a *document-level* semantic representation, while standard truncation would primarily represent the beginning of the article.

5.4.1 Strategy

A chunking strategy was therefore introduced.
For article i, the model input is constructed as

$$X_i = \text{title}_i + \texttt{\textbackslash n\textbackslash n} + \text{body}_i. \tag{5.7}$$

The complete text is tokenized without truncation.
MiniLM supports a maximum sequence length of 256 tokens. Since two positions are reserved for model-specific special tokens, each content chunk contains at most 256-2 = 254 content tokens.
The complete token sequence is therefore partitioned into non-overlapping chunks of 254 content tokens.

5.5 Chunk-Level Embeddings

Suppose article i contains C_i chunks. MiniLM produces one 384-dimensional vector for each chunk:

$$e_{i,c} \in \mathbb{R}^{384}, \qquad c = 1, \ldots, C_i. \tag{5.8}$$

Each chunk embedding is then L_2-normalized:

$$\hat{e}_{i,c} = \frac{e_{i,c}}{|e_{i,c}|_2}. \tag{5.9}$$

This design ensures that each content token contributes once to the document representation. An overlapping strategy would cause tokens appearing in overlap regions to be represented multiple times and would therefore introduce implicit additional weighting.

Zero overlap is thus a simple and reproducible implementation choice. It should be interpreted as part of the document-embedding specification developed for this thesis rather than as an intrinsic requirement of MiniLM.

5.6 Document-Level Mean Pooling

The normalized chunk embeddings are aggregated through arithmetic mean pooling:

$$\bar{e}_i = \frac{1}{C_i} \sum_{c=1}^{C_i} \hat{e}_{i,c}. \tag{5.10}$$

The resulting vector is then L_2-normalized:

$$E_i = \frac{\bar{e}_i}{\|\bar{e}_i\|_2}. \tag{5.11}$$

So that $\|E_i\|_2 = 1$, and the final output is a single 384-dimensional vector for each article.
Mean pooling provides a method for combining information distributed across the complete article while preserving the native dimensionality of the MiniLM embedding space.
Importantly, the dimensionality of 384 was not independently chosen during the thesis. It is the native output dimension of `all-MiniLM-L6-v2`. Chunking and pooling preserve this dimensionality.

5.7 Cosine Similarity and Normalization

For two normalized article embeddings E_i and E_j, cosine similarity is defined as

$$\cos(E_i, E_j) = \frac{E_i^\top E_j}{\|E_i\|_2 \|E_j\|_2}. \tag{5.12}$$

Since the document embeddings are L_2-normalized, with $\|E_i\|_2 = \|E_j\|_2 = 1$, this simplifies to

$$\cos(E_i, E_j) = E_i^\top E_j, \tag{5.13}$$

and is the cosine similarity.
This property is especially useful for the emerging-risk measures developed in the following chapter, where semantic relationships between current and historical news articles are repeatedly evaluated.
High cosine similarity indicates semantically similar articles, while lower cosine similarity indicates greater semantic distance.

5.8 Validation of the Chunking Strategy

The first full-document benchmark was conducted on 1,000 articles.
The mean number of chunks per article was approximately 4166 and only 5.6% of the sampled articles could be represented using a single chunk.
These results confirmed that the long-document issue was not confined to a small number of extreme observations. Multi-chunk processing was necessary for most of the corpus.

A larger audit was subsequently conducted to verify that the initial benchmark was not driven by a particular period of the sample and similar results were obtained. The temporal audit therefore provided additional evidence that the chunking requirements were stable across the sample period.

5.9 Final Embedding Specification

The final document-embedding specification is identified as

`all-MiniLM-L6-v2_chunk254_mean_l2_v1`,

and its defining characteristics are:

- backbone: `sentence-transformers/all-MiniLM-L6-v2`;
- output dimension: 384;
- input: article title followed by article body;
- maximum model sequence length: 256;
- content tokens per chunk: 254;
- chunk overlap: zero;
- chunk-level (L_2) normalization;
- arithmetic mean pooling;
- final document-level (L_2) normalization.

5.10 Full-Corpus Production

The final embedding pipeline was applied to the complete analytical corpus.
The resulting database contains 2,249,255 document embeddings, corresponding one-to-one with the valid analytical articles.
Every vector has dimension 384, and no article in the final corpus has a missing document embedding.
The embeddings are stored at article level and constitute a semantic representation independent of the five predefined risk categories.
This distinction is essential for the next stage of the analysis. The thematic scores identify the degree to which an article is associated with a known risk category, whereas the document embedding describes the article's overall semantic content.

5.11 Role of Embeddings in the Empirical Strategy

The document embeddings are not themselves interpreted as emerging-risk indicators. Instead, they provide the semantic geometry required to construct such indicators.

In particular, the next chapter uses the vectors in two distinct ways: first, article-level novelty is constructed by comparing each article with semantically similar articles observed in the historical information set and second, semantic drift is constructed by aggregating embeddings within theme–month groups and measuring changes in the semantic representation of a risk category over time.

The embedding layer therefore connects unstructured news text to quantitative measures of semantic change:

$$\text{news text} \rightarrow E_i \rightarrow \{\text{semantic novelty, semantic drift}\} \tag{5.14}$$

Chapter 6

Emerging Risk Measures

The previous two chapters developed two complementary article-level representations of the news corpus. First, thematic relevance scores ($S_{i,k}$) quantify the extent to which each article is associated with each of the five predefined risk categories. Second, document embeddings (E_i) provide a general semantic representation of the information contained in each article.

These representations provide the inputs required to introduce the temporal dimension of the analysis. The objective of this chapter is to transform article-level textual information into quantitative measures capable of describing different manifestations of financial risk emergence. A central premise of the empirical framework is that emergence is not a single directly observable phenomenon. A risk may begin to emerge because new information appears that is substantially different from previously observed news. The meaning and mechanisms associated with an existing risk category may change over time. A risk may receive an increasing share of media attention. Finally, attention may increase sufficiently rapidly and materially to constitute an exceptional attention episode rather than an ordinary fluctuation in coverage. The empirical framework therefore distinguishes four complementary dimensions:

1. **Coverage**, measuring the relative amount of news attention devoted to a risk category;
2. **Novelty**, measuring whether the semantic information contained in an article is unusual relative to recent historical news;
3. **Semantic Drift**, measuring whether the semantic representation associated with a risk category changes over time;
4. **Burst Detection**, identifying periods in which thematic attention experiences an unusually sharp and material increase relative to its own historical dynamics.

These measures capture distinct dimensions of the information environment and should not be interpreted as alternative estimates of the same latent quantity. Novelty is initially defined at the article level. Semantic drift and coverage are constructed at the theme–month level. Burst detection is subsequently applied to the monthly coverage series. The resulting analytical sequence can be summarized as

$$\boxed{\text{Thematic Scores + Document Embeddings + Time} \longrightarrow \text{Emerging Risk Measures}}. \tag{6.1}$$

6.1 Coverage

« How much coverage does a particular risk receive in the US press each month? »

The first indicator built in this thesis is a more sophisticated version of the one in Cherubini et al., 2024. Coverage measures the presence of each risk in the monthly news flow.

6.1.1 Definition and Choice of the Coverage Measure

For theme k and month t, let N_t denote the total number of valid articles published during the month. Two alternative coverage measures were considered.
The first uses the thematic scores directly:

$$C_{k,t}^{weighted} = \frac{\sum_{i \in t} s_{i,k}}{N_t}. \tag{6.2}$$

This measure combines two components: the number of related articles and the strength of their thematic scores.
The second measure uses positive thematic evidence:

$$C_{k,t}^{binary} = \frac{\#\{i \in t : s_{i,k} > 0\}}{N_t}. \tag{6.3}$$

It has a direct interpretation as the share of valid monthly articles with positive evidence for risk theme k.
The two definitions were compared before selecting the production specification. Table 6.1 reports their monthly correlations for each risk category.

Table 6.1: Correlation between binary and weighted monthly coverage

Risk Theme	Correlation
Cyber Risk	0.976537
Climate Risk	0.998625
Pandemic Risk	0.999777
Supply Chain Risk	0.998478
AI-Related Risk	0.996814

The correlations are high for all five risk categories. The two definitions therefore produce very similar temporal patterns.
The binary-positive measure was selected for final production. Its main advantage is its simple interpretation as a share of the monthly news flow. It also avoids giving additional importance to differences in the magnitude of thematic scores.
Let

$$N_{k,t} = \#\{i \in t : s_{i,k} > 0\} \tag{6.4}$$

denote the number of articles with positive evidence for theme k in month t. Final monthly coverage is then defined as

$$\boxed{x_{k,t} = C_{k,t} = \frac{N_{k,t}}{N_t}}. \tag{6.5}$$

The measure ranges from zero to one. Higher values indicate that a larger share of the monthly news flow is associated with the risk category.
The normalized coverage series $x_{k,t}$ is then used as the input to the burst-detection procedure developed in Section 6.4.

6.1.2 Coverage Results

The sample contains 120 months, from January 2014 to December 2023. With five risk categories, the resulting panel contains 600 theme-month observations.
Table 6.2 reports positive thematic observations and the main coverage statistics for each category.

Table 6.2: Monthly coverage by risk theme

Theme	Total Positive	Min. Monthly	Max. Monthly	Mean Coverage	Max. Coverage
Cyber Risk	38,664	110	715	0.017405	0.032366
Climate Risk	120,017	477	1,948	0.059039	0.128928
Pandemic Risk	195,075	0	11,974	0.108873	0.753556
Supply Chain Risk	47,787	143	1,169	0.024822	0.076560
AI-Related Risk	6,484	4	214	0.003529	0.018551

Coverage is calculated for all 600 theme-month combinations. The findings further reveal significant variation in the presence of the five risks in the news.
Pandemic risk has the highest mean coverage of 0.108873 and a maximum of 0.753556. It has a monthly count ranging between zero and 11,974 positive articles. The broad range is due to the high concentration of news about pandemic risk in part of the data set.
Climate risk comes in second place in terms of mean coverage, which is 0.059039. Supply-chain and cyber risks have mean coverage of 0.024822 and 0.017405, respectively. The risk associated with AI has the lowest mean coverage of 0.003529.
A zero value is a legitimate coverage observation. When there is not a single article in month t with positive evidence of theme k, then

$$C_{k,t} = 0. \tag{6.6}$$

This treatment differs from semantic drift. When thematic support is insufficient for a reliable semantic representation, the corresponding drift observation is missing rather than zero:

$$D_{k,t} = \text{NA}. \tag{6.7}$$

The distinction is important. Zero coverage indicates an observed absence of positive thematic evidence in the monthly news flow. Missing semantic drift instead indicates that semantic change cannot be estimated reliably.

6.2 Novelty

« To what extent is an article semantically new compared with the news that was already available before its publication? »

Measuring the novelty of a text as a distance from previously observed texts has its origins in information retrieval. Allan et al., 1998 and Brants and Chen, 2003 address the problem of new event detection: identifying, within a news stream, stories concerning an event that has never been seen before. Brants and Chen solve this using an incremental TF-IDF model, which compares each new story with those already present in the stream. In economics, Kelly et al., 2021 apply a similar principle to patents. They compare the text of each patent with that of preceding and subsequent patents, and identify as relevant those that differ from the preceding ones but are similar to the subsequent ones. More recently, Humann et al., 2026 use paragraph-level embeddings on FOMC minutes. They assign each paragraph a semantic distance from the closest paragraph in previous communications, aggregate the scores into novelty indices, and link these to financial market reactions.

This chapter follows the same logic, but applies it to a corpus of over two million news articles. Instead of the single most similar neighbour, novelty is calculated based on the ten closest historical articles within a 365-day window. The neighbour search is made possible by HNSW (Malkov and Yashunin, 2020), a graph-based approximate neighbour search method, implemented using the FAISS library (Johnson et al., 2021).

The objective of semantic novelty is to determine whether the current article has information which has been already mentioned in past articles. Otherwise, the article introduces new information from the semantic viewpoint. As all the articles have a normalized document embedding associated with them, novelty can be expressed as a local distance in the semantic space. If the article resembles past articles, then much of the information contained in it would already be there in the past news. Hence, it will have low novelty. On the other hand, if the article dissimilar to past articles, then much of the information in it would have less presence in past news. Therefore, the novelty of the article will be high. Thus, the measurement is dependent on the semantic neighborhood of each article rather than their absolute position in the embedding space.

6.2.1 Historical Semantic Neighbourhood

For each article i, a historical candidate set is defined as

$$\mathcal{H}_i(W) = \left\{ j : t_i - W \leq t_j < t_i \right\}, \tag{6.8}$$

where t_i denotes the publication date of article i and W is the historical look-back window. Within this historical information set, the K articles with the highest cosine similarity to the current article are identified. Let

$$NN_K(i; W) \tag{6.9}$$

denote the set of these K nearest historical semantic neighbours. The general novelty measure is

$$N_i(K, W) = 1 - \frac{1}{K} \sum_{j \in NN_K(i;W)} \cos(E_i, E_j). \tag{6.10}$$

Since all document embeddings are L_2-normalized,

$$\cos(E_i, E_j) = E_i^\top E_j. \tag{6.11}$$

The measure can therefore be interpreted as one minus the average semantic similarity between the current article and its closest historical neighbours.
A high average similarity produces a low novelty score and a lower average similarity produces a higher novelty score.

Novelty is a semantic-distance measure rather than a probability. A value of $N_i = 0.20$, for example, should not be interpreted as implying that an article is "20% new".

6.2.2 Choice of the Number of Neighbours and Historical Window

The novelty measure depends on two main parameters: the number of nearest neighbours K and the historical window W. Both parameters were selected through a calibration exercise on the news corpus. The aim was not to maximize or minimize novelty. Instead, the calibration looked for a balance between historical coverage, semantic locality, and score stability.
Three historical windows were tested:

$$W \in \{30, 90, 365\} \text{ days}, \tag{6.12}$$

together with five values for the number of neighbours:

$$K \in \{1, 5, 10, 20, 50\}. \tag{6.13}$$

The resulting grid contained 15 alternative specifications. The pilot used 1,080 target articles, producing 16,200 article-specification observations.

For each specification, the candidate set included only articles within the selected historical window:

$$d_i - W \leq d_j < d_i. \tag{6.14}$$

The strict inequality means that only information published before the target article can enter its reference set. Articles from the same calendar date are excluded.
This restriction is important because reliable intraday publication times are not available for all articles. A same-day article could have been published after the target article. Including it could therefore introduce look-ahead contamination into the historical benchmark.

The parameter W controls how much past information is available when searching for semantic neighbours. As expected, longer windows produced much larger candidate sets. With $W = 30$, the

median candidate set contained about 16,090 articles. It increased to 47,766 for $W = 90$ and 204,664 for $W = 365$.

The window length also affected the level of measured novelty. With $K = 10$, mean novelty was 0.330 for the 30-day window. It fell to 0.297 for 90 days and 0.265 for 365 days.

This behavior has a simple explanation. A longer window gives the model more chances to find a relevant historical precedent. With a shorter window, related information may fall outside the reference period and appear new.

However, using the entire historical corpus would create a different issue. Very old articles could affect whether current information is considered novel. The measure would then become less connected to the recent information environment.

For these reasons, $W = 365$ was selected as the baseline. It provides one year of historical information while keeping the reference period clearly defined. This choice is part of the methodology and is not imposed by the embedding model.

The 365-day window also creates a boundary condition at the start of the sample. The corpus begins in 2014. Articles from that year do not have a complete year of previous observations.

Therefore, 2014 is used as a warm-up period for the novelty measure for the main analytical period runs from 2015 to 2023. This prevents missing pre-sample information from being interpreted as unusually high semantic novelty.

The parameter K plays a different role. It determines how many historical articles form the local semantic benchmark.

With $K = 1$, novelty depends only on the closest historical article. The score can therefore be sensitive to a near-duplicate or one unusually similar story.

A large K creates the opposite issue. It gradually includes less similar articles, making the benchmark broader but less local in semantic terms.

Table 6.3 summarizes the main calibration results. Panel A compares the three historical windows while keeping $K = 10$. Panel B compares alternative values of K within the 365-day window.

Table 6.3: Novelty calibration results

Panel A: Historical window ($K = 10$)

W (days)	K	Mean novelty	Spearman correlation
30	10	0.330	0.884
90	10	0.297	0.945
365	10	0.265	1.000

Panel B: Number of neighbours ($W = 365$)

W (days)	K	Mean novelty	Spearman correlation
365	1	0.210	0.880
365	5	0.245	0.988
365	10	0.265	1.000
365	20	0.287	0.990
365	50	0.320	0.953

Note: Spearman correlations are computed against the final specification, $K = 10$ and $W = 365$ days.

Panel A shows that mean novelty falls as the historical window becomes longer. A larger reference set makes it easier to find relevant historical precedents. Article rankings remain fairly stable across the three windows.
Panel B shows what happens when the number of neighbours changes. Values close to $K = 10$ produce very similar article rankings. The Spearman correlation is 0.988 for $K = 5$ and 0.990 for $K = 20$. It falls to 0.953 when K increases to 50.

Overall, the results support a broad historical window combined with a relatively local semantic neighbourhood. The final specification therefore uses

$$\boxed{K = 10} \tag{6.15}$$

and

$$\boxed{W = 365 \text{ days}}. \tag{6.16}$$

The choice of $K = 10$ balances stability and semantic locality. The score does not depend on one historical article, while the neighbourhood remains relatively small.
The 365-day window limits false novelty caused by a short historical memory. At the same time, the comparison remains linked to a recent information environment.

Importantly, financial-market outcomes were not used to select K or W. The calibration only considered the behaviour and stability of the novelty measure. Both parameters were therefore fixed before the later market analysis.
Given these choices, the final article-level novelty measure is

$$N_i = 1 - \frac{1}{10} \sum_{j \in NN_{10}(i;365)} \cos(E_i, E_j), \tag{6.17}$$

where $NN_{10}(i; 365)$ contains the ten most similar historical articles from the previous 365 days, subject to $d_j < d_i$.

6.2.3 Computational Implementation

The corpus contains more than two million document embeddings, each with 384 dimensions. Comparing every article with all eligible historical articles would therefore require a very large number of computations.
To make the process feasible, nearest-neighbour search was implemented using FAISS/HNSW. This method allows similar historical articles to be identified efficiently across the full corpus.
The computational method should be kept separate from the definition of novelty itself. Equation (6.17) defines how the novelty score is calculated. FAISS/HNSW is instead used to find the relevant historical neighbours efficiently.
The resulting novelty scores are stored at article level in

`risk_emergence.news_novelty`.

Novelty is therefore first computed for individual articles. It remains distinct from the monthly theme-level indicators introduced in the following sections.

6.3 Semantic Drift

« To what extent is the semantic content of this risk changing over time? »

Changes in meaning over time are studied primarily in computational linguistics. Hamilton et al., 2016 measure semantic change using embeddings constructed separately for different time periods and then aligned with one another. They assess their reliability against already known historical changes and derive two statistical laws governing changes in meaning. In the field of finance, Hoberg and Hoberg and Phillips, 2016 construct measures of similarity between companies based on product descriptions in 10-K filings. The measures are calculated year by year and therefore vary over time. This work is less directly relevant to our case, as it compares companies rather than the meaning of a category, but it demonstrates how text-based measures that vary over time are already used in the financial literature.

This chapter applies a similar idea to a risk category. The monthly semantic centroid, weighted by thematic scores, is compared with the average of the centroids from the previous twelve months.

A risk label may remain unchanged while the events, technologies, institutions, or consequences linked to that risk evolve. Coverage alone cannot capture this change. More articles indicate greater attention, but they do not show whether the content of that attention is also changing.
Semantic drift addresses this issue through a monthly semantic representation of each risk category. This representation is compared with its recent historical state. The procedure requires three main choices. These concern the representation of each theme, the selection of reliable observations, and the construction of the historical benchmark.

6.3.1 Combining Thematic Scores and Document Embeddings

The theme-specific semantic representation combines two article-level outputs developed in the previous chapters. The document embedding E_i represents the semantic content of article i. The thematic score $s_{i,k}$ measures the strength of its association with risk category k.
For semantic drift, the thematic signal is the frozen weakly supervised continuous score:

$$s_{i,k} = S^{WS}_{i,k}. \tag{6.18}$$

Only articles with positive thematic evidence are included:

$$s_{i,k} > 0. \tag{6.19}$$

For each theme k and month t, these articles are combined into a score-weighted semantic centroid:

$$\tilde{c}_{k,t} = \frac{\sum_{i \in (k,t): s_{i,k}>0} s_{i,k} E_i}{\sum_{i \in (k,t): s_{i,k}>0} s_{i,k}}. \tag{6.20}$$

The thematic score determines the contribution of each article. Articles with stronger thematic evidence therefore receive greater weight.
The resulting vector is then L_2-normalized:

$$c_{k,t} = \frac{\tilde{c}_{k,t}}{\|\tilde{c}_{k,t}\|_2}. \tag{6.21}$$

The normalized centroid $c_{k,t}$ represents the semantic position of risk category k in month t.

6.3.2 Choice of the Thematic Signal

The thematic signal was selected by comparing the empirical properties of the available scoring methods. The comparison focused mainly on the weakly supervised and supervised approaches.
On the independent final test, the weak continuous method achieved

$$\text{ROC-AUC}_{WS} = 0.785135, \tag{6.22}$$
$$\text{AP}_{WS} = 0.583126, \tag{6.23}$$

whereas the supervised V3 model achieved

$$\text{ROC-AUC}_{SUP} = 0.742859, \tag{6.24}$$
$$\text{AP}_{SUP} = 0.112327. \tag{6.25}$$

The weak score showed stronger ranking performance, especially in terms of Average Precision. It was therefore selected as the thematic signal for semantic drift.
However, the weak signal is highly sparse. About 95.8% of its observations in the final test were equal to zero. For this reason, the reliability of monthly semantic representations could not be assumed. A separate selection and validation procedure was needed.

6.3.3 Selection Strategy and Reliability Validation

Several selection rules were tested before defining the final monthly centroids.
The first pilot used a common absolute threshold:

$$S_{i,k,m} \geq 0.5. \tag{6.26}$$

The rule was applied to all three thematic-scoring methods. It generated 1,736 of the 1,800 potential method–theme–month centroids.
However, the three methods produce scores through different mechanisms and have different empirical distributions. A value of 0.5 does not necessarily indicate the same level of thematic evidence across methods. The common absolute threshold was therefore rejected.
A second pilot used a relative selection rule. Within each method–theme–month group, the top 10% of observations were selected by thematic score. This avoided imposing the same numerical threshold on different score distributions.

However, this approach created a problem for the weakly supervised signal. In 54 weakly supervised theme–month groups, all observations in the top decile had a score of zero. In these cases,

$$\sum_i s_{i,k} = 0, \tag{6.27}$$

so a score-weighted centroid could not be constructed.

The problem was also conceptual. If all scores are zero, a fixed percentile selects articles through arbitrary ties. Yet the model provides no positive thematic evidence for those articles. The top-decile approach was therefore rejected.

The final strategy uses the positive support of the weak continuous score:

$$s_{i,k} > 0. \tag{6.28}$$

This rule includes only articles for which the model provides positive thematic evidence. It also avoids an additional absolute or percentile threshold.

The amount of positive evidence varies across risk categories. Cyber, climate, and supply-chain risk show positive thematic evidence throughout the sample. Pandemic risk is very sparse before 2020, with 54 months containing no positive observations. AI-related risk also has less positive support than the more established categories.

A monthly centroid should therefore not be considered reliable simply because one positive article exists. For this reason, a minimum-support requirement was evaluated empirically.

A preliminary analysis showed that consecutive monthly centroids became more similar as thematic support increased. However, this comparison cannot separate estimation noise from genuine semantic change. A split-half reliability procedure was therefore used.

For each theme–month group, positive observations were divided deterministically into two subsets. An independent score-weighted centroid was then constructed for each subset:

$$c_{k,t}^{(A)} \qquad \text{and} \qquad c_{k,t}^{(B)}. \tag{6.29}$$

Their cosine similarity was calculated as

$$R_{k,t} = \cos\left(c_{k,t}^{(A)}, c_{k,t}^{(B)}\right). \tag{6.30}$$

Both centroids describe the same risk category in the same month. Low similarity therefore suggests an unstable semantic representation rather than genuine temporal drift.

The validation used ten deterministic splits. For groups with at least ten positive observations, overall reliability was

$$\text{mean}(R_{k,t}) = 0.966312, \tag{6.31}$$

$$\text{median}(R_{k,t}) = 0.985228, \tag{6.32}$$

$$p05(R_{k,t}) = 0.864616. \tag{6.33}$$

Reliability increased clearly with thematic support, as shown in Table 6.4.

These results were then used to choose the minimum level of thematic support. With

Table 6.4: Split-half centroid reliability by thematic support

Positive articles	Mean	5th percentile	Minimum
10–19	0.774	0.678	0.644
20–29	0.847	0.799	0.781
30–49	0.888	0.830	0.813
50–99	0.928	0.900	0.878
100–299	0.972	0.955	0.916
300–999	0.988	0.981	0.978
≥ 1000	0.996	0.993	0.992

$$n_{\text{positive}} \geq 10, \tag{6.34}$$

520 of the 600 potential theme–month centroids would be retained. However, minimum split-half reliability would be only about 0.644.

Higher thresholds improve reliability but reduce temporal coverage. This trade-off is especially relevant for sparse categories such as AI-related risk.

At

$$n_{\text{positive}} \geq 50, \tag{6.35}$$

461 of the 600 potential centroids are retained. The corresponding reliability statistics are

$$\text{mean}(R) = 0.980656, \tag{6.36}$$

$$p05(R) = 0.932592, \tag{6.37}$$

$$\min(R) = 0.878467. \tag{6.38}$$

A threshold of 100 would further improve average reliability. However, it would remove a substantial number of observations for AI-related risk. The final specification therefore uses

$$\boxed{n_{\text{positive}} \geq 50}. \tag{6.39}$$

This threshold provides a balance between semantic reliability and temporal coverage.

An effective-sample-size diagnostic was also computed:

$$N_{\text{eff}} = \frac{(\sum_i s_i)^2}{\sum_i s_i^2}. \tag{6.40}$$

This statistic helps identify groups dominated by a small number of articles with high weights. The diagnostic did not support an additional production threshold. The final specification therefore retained the simpler $n_{\text{positive}} \geq 50$ rule.

6.3.4 Historical Reference and Semantic Drift

After constructing a reliable current centroid, semantic drift requires a historical benchmark. Three alternatives were considered.

The first option was a month-on-month comparison:

$$D_{k,t}^{MoM} = 1 - \cos(c_{k,t}, c_{k,t-1}). \tag{6.41}$$

This measure is easy to interpret but has two limitations. It cannot be computed when the previous month lacks a reliable centroid. It can also be sensitive to short-term variation.
The second option was a fixed historical baseline:

$$D_{k,t}^{Fixed} = 1 - \cos(c_{k,t}, c_{k,t_0}). \tag{6.42}$$

This approach depends strongly on the choice of t_0. As time passes, it also measures increasing displacement from a single historical state.
The final specification uses a rolling historical reference. For each current month t, the reference can use reliable centroids from the previous 12 calendar months:

$$\mathcal{H}_{k,t} = \left\{ c_{k,\tau} : 1 \leq t - \tau \leq 12 \right\}. \tag{6.43}$$

Only reliable historical centroids are included. Missing months are not imputed. Current and future information never enters the historical reference.
Let

$$h_{k,t} = |\mathcal{H}_{k,t}| \tag{6.44}$$

denote the number of available historical centroids. At least three are required:

$$h_{k,t} \geq 3. \tag{6.45}$$

The historical monthly centroids receive equal weight. Their unnormalized mean is

$$\tilde{r}_{k,t} = \frac{1}{h_{k,t}} \sum_{c_{k,\tau} \in \mathcal{H}_{k,t}} c_{k,\tau}. \tag{6.46}$$

The historical reference is then L_2-normalized:

$$r_{k,t} = \frac{\tilde{r}_{k,t}}{\|\tilde{r}_{k,t}\|_2}. \tag{6.47}$$

The two weighting stages therefore serve different purposes. Articles within a theme–month are weighted by their thematic scores. Historical monthly centroids instead receive equal weight. As a result, months with more articles do not automatically dominate the historical benchmark.
The final semantic-drift measure is

$$\boxed{D_{k,t} = 1 - \cos(c_{k,t}, r_{k,t})}. \tag{6.48}$$

Higher values indicate greater semantic displacement from the recent historical representation of the same risk category.
Missing observations are kept separate from zero semantic drift. A reliable current centroid is not constructed when a theme–month contains fewer than 50 positive articles. Semantic drift is also unavailable when fewer than three reliable historical centroids exist in the previous 12 months.

In either case,

$$D_{k,t} = \text{NA}. \tag{6.49}$$

A missing value indicates that the available information is insufficient to estimate semantic change reliably. This differs from $D_{k,t} = 0$. A zero value means that a valid current representation coincides with its historical benchmark.

6.3.5 Production Results

The final production procedure processed 408,027 positive article–theme observations. Positive thematic mass was present in 546 of the 600 possible theme–month combinations.
After the minimum-support rule was applied, 461 reliable monthly centroids remained. The rolling historical-reference requirements then produced 443 valid semantic-drift observations.
Table 6.5 reports the distribution of the final measure.

Table 6.5: Distribution of the final semantic-drift measure

Statistic	Semantic Drift
Minimum	0.006654
Mean	0.029467
Median	0.023402
95th percentile	0.067083
Maximum	0.154270

The final semantic-drift values were stored in the monthly theme-level indicator structure.

6.4 Burst Detection

« Is the current news volume unusually high compared to its historical baseline? »

Bursts in text streams are formalised by Kleinberg, 2003. He considers the stream as an infinite-state automaton in which bursts are state changes and uses the technique to study archives of emails and scientific articles. In finance, Barber and Odean, 2008 find that individual investors prefer to purchase securities that receive attention. Attention is measured via news coverage, unusual trading volume and abnormal daily return of the security. Da et al., 2011 measure attention to a stock directly through the number of Google searches and use abnormal attention as the primary variable. Manela and Moreira, 2017 build a textual index from the front page of the Wall Street Journal and document peaks at market crashes, wars and financial crises.

In this chapter, I extend this literature by measuring media attention to a risk as a time series and comparing it to its recent past. In contrast to the cited papers, the measure involves a 12-month z-score, a materiality threshold defined by the 95th historical percentile and persistence of the abnormal value. The values of these thresholds are the calibration parameters for this chapter, selected based on the sensitivity analysis in Section 6.4.3.

Coverage measures the amount of news attention linked to each risk category. However, it does not separate normal variation from unusually sharp increases in attention.
Burst detection adds this second dimension. Its purpose is to identify exceptional increases relative to the recent history of each risk. The method also separates the onset of an attention shock from its persistence over time.

6.4.1 Historical Benchmark

Burst detection is applied to the normalized monthly coverage series $x_{k,t}$, defined in Equation (6.5). Normalized coverage is used instead of raw article counts because the total volume of news changes over time.
For each theme, the historical benchmark is based on the previous 12 months. The rolling mean is

$$\mu_{k,t}^{(12)} = \frac{1}{12} \sum_{j=1}^{12} x_{k,t-j}, \tag{6.50}$$

while the corresponding historical standard deviation is

$$\sigma_{k,t}^{(12)} = \mathrm{sd}\left(x_k, t-12, \ldots, x_k, t-1\right). \tag{6.51}$$

The current observation $x_{k,t}$ is excluded from both statistics. The benchmark is therefore strictly backward-looking. Information from month t cannot affect the historical distribution used to assess that same month.
A standardized coverage shock is then calculated as

$$Z_{k,t}^{(12)} = \frac{x_{k,t}\mu^{(12)}k,t}{\sigma^{(12)}k,t}. \tag{6.52}$$

This statistic measures how unusual current coverage is relative to the recent variability of the same risk category.

6.4.2 Burst Onset, Persistence, and Intensity

A high value of $Z_{k,t}^{(12)}$ is not enough on its own to identify a burst onset. When historical coverage is very low, even a small absolute increase can generate a large standardized deviation.
For this reason, the onset rule also includes a historical materiality condition. Let

$$Q_{0.95,k,t}^{\mathrm{hist}} = Q_{0.95}\left(x_{k,1}, \ldots, x_{k,t-1}\right) \tag{6.53}$$

denote the empirical 95th percentile of all coverage observations available for theme k before month t. This percentile uses the full available history of the theme rather than only the 12-month rolling window.
If the theme is not already in an active burst, a new episode begins when

$$\boxed{Z_{k,t}^{(12)} > 3 \qquad \text{and} \qquad x_{k,t} > Q_{0.95,k,t}^{\mathrm{hist}}}. \tag{6.54}$$

The two conditions have different roles. The z-score identifies an unusual increase relative to recent dynamics. The historical percentile requires coverage to also be high relative to the broader history of

that theme. This filter reduces signals caused by large standardized changes from very low coverage levels.
Burst detection is implemented as a state process. Once an episode starts, the onset conditions are not applied again in every following month. An active burst continues while

$$x_{k,t} > \mu_{k,t}^{(12)}. \tag{6.55}$$

The episode ends in the first month for which

$$x_{k,t} \leq \mu_{k,t}^{(12)}. \tag{6.56}$$

Any new onset signal observed during an active episode is absorbed into the existing burst. It is therefore not counted as a separate episode.
The resulting burst-state variable is

$$B_{k,t} = \begin{cases} 1 & \text{if theme } k \text{ is in an active burst in month } t, \\ 0 & \text{otherwise.} \end{cases} \tag{6.57}$$

The binary state shows whether a burst is active, but it does not measure its size. A separate intensity variable is therefore defined as

$$I_{k,t} = \begin{cases} x_{k,t} - \mu^{(12)} k, t & \text{if } Bk, t = 1, \\ 0 & \text{if } B_{k,t} = 0. \end{cases} \tag{6.58}$$

The onset statistic and the intensity measure therefore capture two different aspects of a burst. The z-score detects unusual changes relative to recent variability. Burst intensity instead measures excess attention in units of normalized coverage.
This distinction matters when historical variability is very small. In these cases, standardized shocks can become very large even when the absolute change in coverage is limited.

6.4.3 Specification and Calibration

The final specification was selected through a set of diagnostic exercises before storing the burst indicators in the final dataset.
Several alternatives were considered. They included different historical windows, robust dispersion measures, count-based and binomial formulations, alternative materiality filters, and different state rules. Frozen historical benchmarks and alternative measures of burst intensity were also examined.
A 12-month backward-looking window was retained for the historical benchmark. It gives a local comparison while still covering a full annual period.
For the onset condition, standardized thresholds of 2.5, 3.0, and 3.5 were compared. The final specification uses

$$\boxed{Z_{k,t}^{(12)} > 3}. \tag{6.59}$$

The lower threshold generated additional weak episodes. In contrast, the higher threshold removed or delayed several major increases in thematic attention. A threshold of three standard deviations gave a more balanced result.

The all-prior 95th-percentile filter was retained as the materiality condition. During calibration, it removed standardized spikes linked to very low coverage levels. At the same time, it preserved the main attention episodes identified by the rolling (z)-score.
Alternative persistence rules were also tested. One option was to freeze the historical benchmark when a burst started. However, this rule produced overly persistent episodes when thematic coverage moved to a more lasting level.
The final specification instead updates the 12-month benchmark every month. This allows the burst state to adjust gradually when the underlying attention environment changes.
Alternative intensity measures were considered as well. One option was relative excess coverage,

$$\frac{x_{k,t}}{\mu_{k,t}^{(12)}} - 1, \tag{6.60}$$

which becomes unstable when historical coverage is close to zero. A bounded alternative was

$$\frac{x_{k,t} - \mu^{(12)} k, t}{xk, t}. \tag{6.61}$$

This measure avoids the same instability, but it compresses large attention episodes. For this reason, the absolute-excess measure in Equation (6.58) was retained.
The final specification separates detection from measurement. Onset depends on a standardized shock together with a historical materiality filter. Persistence requires coverage to remain above its moving historical mean. Intensity then measures the absolute excess above that benchmark.

6.4.4 Warm-Up Period

The rolling benchmark requires 12 prior monthly observations. Therefore, the first 12 months available for each theme form a warm-up period.
During these months,

$$B_{k,t} = I_{k,t} = \text{NA}, \qquad t = 1, \ldots, 12. \tag{6.62}$$

A missing burst state is different from a state equal to zero. A value of $B_{k,t} = 0$ means that the month was evaluated and no active burst was found. A missing value instead means that the required historical information was not yet available.

6.4.5 Production Results

The procedure was applied to the five risk categories over the 2014–2023 sample. The complete panel contains 600 theme–month observations.
For each theme, the first 12 months form the warm-up period. This gives 60 non-evaluable observations and leaves 540 evaluable theme–month observations. Of these, 110 are classified as active burst months and 430 as inactive months. Overall, the procedure identifies 19 distinct burst episodes.
Table 6.6 reports the distribution of the final burst indicators across the five risk categories.
The results show that burst frequency and burst magnitude capture different features of news attention. Supply-chain risk records the most active burst months, with 35 months across five episodes. AI-related risk instead has the largest number of distinct episodes, with six episodes covering 26 months.

Table 6.6: Burst-detection results by risk theme

Theme	Active Burst Months	Episodes	Maximum Burst Intensity
Cyber Risk	15	3	0.01310225
Climate Risk	22	4	0.05103656
Pandemic Risk	12	1	0.68381492
Supply Chain Risk	35	5	0.04025854
AI-Related Risk	26	6	0.01165945
Total	110	19	–

Pandemic risk follows a different pattern. It has only one burst episode, which lasts 12 months, but reaches a maximum intensity of 0.683815. This value is substantially higher than the maximum intensity recorded for any other theme. It reflects the exceptional rise in pandemic-related news coverage during 2020.
Cyber risk records 15 active burst months across three episodes. These occur in December 2018–November 2019, January 2020, and February–March 2022. They provide the burst component of the cyber-risk indicators used in the later empirical analysis.
The final values were stored in the monthly theme-level indicator dataset as `burst_state` and `burst_intensity`. The database write was checked against the final pre-commit output. All 600 theme–month observations matched, with no differences in either burst state or burst intensity.

6.5 Joint Interpretation of the Emerging Risk Measures

The four metrics introduced in this chapter each measure a different aspect of financial risk emergence. They measure changes in the information content, meaning of the topic, prevalence of the news, and dynamics of attention.

Novelty is an article-level metric. It measures whether an article contains semantic information that is different from the recent history of news on the topic. Therefore, a high novelty score means the appearance of relatively novel information. It does not mean that the related risk is well-recognized by media.

Semantic drift is a theme-month metric. It measures whether the recent semantic representation of a risk category differs from its historical representation. Therefore, a risk can have semantic drift even if the name of the risk category stays the same. The metric measures changes in the events, mechanism, and consequences of the risk category.

Coverage is a different metric. It measures the proportion of monthly news that contain positive evidence about a given risk category. More coverage means more thematic prevalence in the news flow. But it does not say whether the information is new or whether the attention increase is unusual.

Burst detection adds this final distinction. It identifies unusually sharp and material increases in coverage relative to the theme's own historical dynamics. Burst intensity then measures the amount of excess coverage during an active episode. A risk can therefore have high coverage without being in a

burst if that level of attention is not exceptional relative to its recent history.

The conceptual distinction between the four measures is summarized in Table 6.7.

Table 6.7: Interpretation of the emerging-risk measures

Measure	Main Question
Novelty	Is the information in the article semantically new relative to recent historical news?
Semantic Drift	Is the meaning associated with the risk category changing relative to its recent historical representation?
Coverage	How much of the current news flow is associated with the risk category?
Burst Detection	Is thematic attention increasing in an unusually sharp and material way relative to its own history?

Therefore, the measures can be seen as complementary, and not as substitutes. Different combinations can characterize different modes of emergence of risks.
For instance, high level of coverage along with low semantic drift characterizes risks which attract a lot of attention but at the same time remain quite stable in terms of their semantic content. High semantic drift and low level of coverage indicate a change of meaning in a risk but its low prevalence in the news flow.
Another distinction is provided by novelty. New semantics can emerge even before a certain risk becomes prevalent in the media. Thus, a novelty can increase without an increase in the coverage immediately. On the other hand, high coverage can be explained by discussions of familiar information and thus coincide with low novelty.
Finally, coverage and burst detection also need to be distinguished. Persistent high levels of coverage do not mean an emergence of a new shock. For detecting bursts, it is necessary to have an extraordinary increase of a topic as compared to its previous historical behaviour.

Thus, altogether, four indicators can characterize different aspects of emergence of financial risks: changes of information content, changes of meaning, prevalence of risks

In summary, the four measures constitute a multidimensional measure of the emergence of financial risk. Changes in information content, theme, prevalence, and dynamics of attention are separated. The news-based indicators generated form the foundation for the empirical analysis that follows.

Chapter 7

Empirical Analysis of Emerging Risk Measures

The previous chapter introduced four complementary measures designed to capture different dimensions of risk emergence in financial news: Novelty, Semantic Drift, News Coverage, and Burst Attention. The present chapter examines how they behave empirically over time and in the five risk themes considered in the study.

Particular attention is paid to the temporal dimension of the measures. Risk emergence is inherently dynamic: a theme may receive sustained attention without generating substantially new information, while another may experience a sudden change in semantic content before becoming widely discussed. Examining the indicators over time therefore allows the analysis to distinguish between changes in informational novelty, semantic content, news attention, and abnormal increases in attention.

Special attention is paid to Cyber Risk.

7.1 Coverage

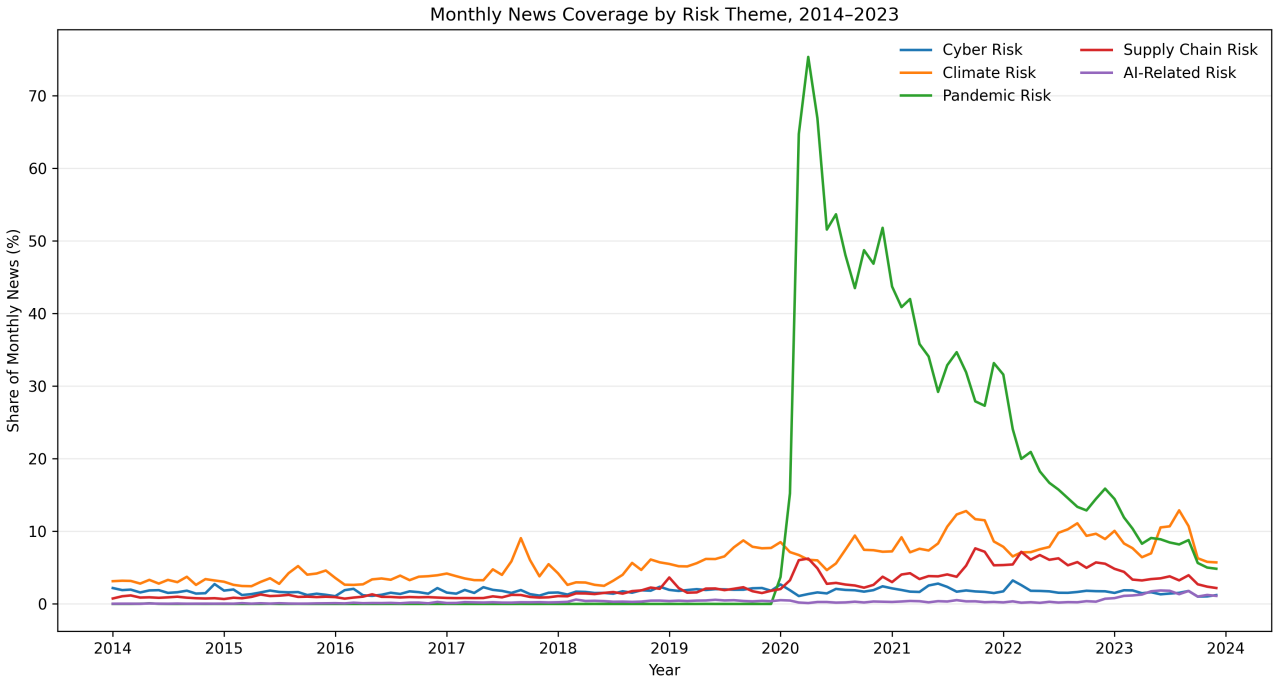


Figure 7.1: Monthly News Coverage by Risk Theme, 2014–2023

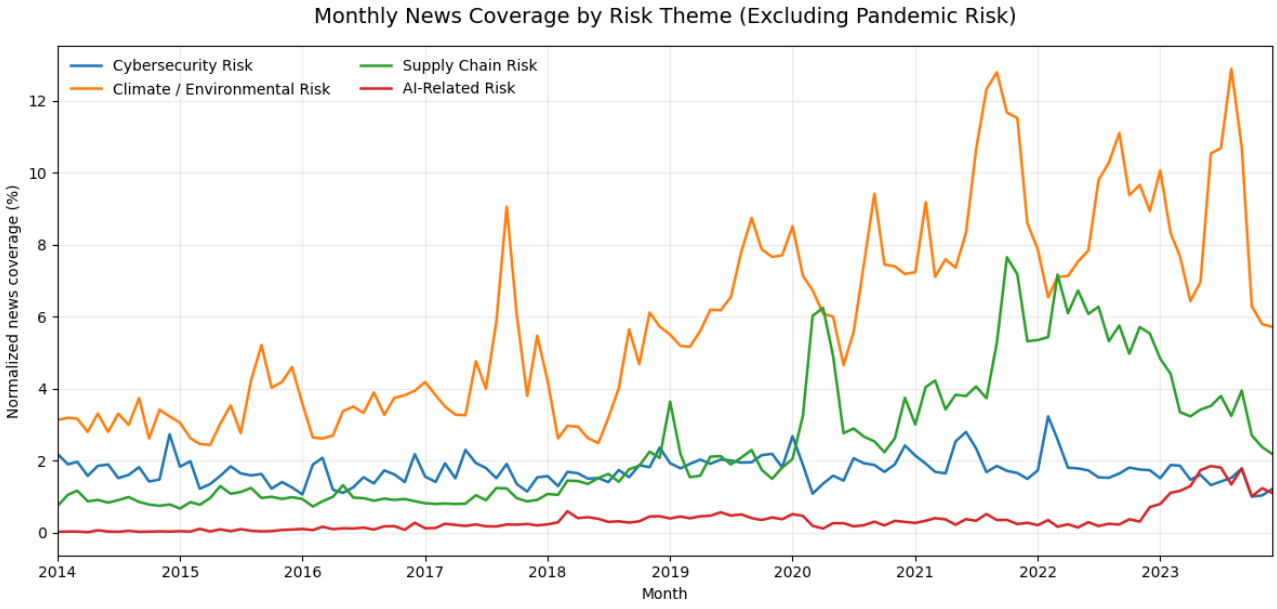


Figure 7.2: Monthly News Coverage by Risk Theme, 2014–2023

Figure 7.1 shows the evolution of News Coverage for the five risk themes between 2014 and 2023. The differences across risks are large, both in terms of their overall level of attention and in the way this attention changes over time.

Pandemic Risk shows by far the largest change in Coverage. Before 2020, its presence in the news is very limited. This changes abruptly at the beginning of 2020, when Coverage rises to more

than 70% of monthly news at its highest point. Attention remains very high during the rest of 2020 and 2021, although it gradually declines. The downward trend continues through 2022 and 2023. By the end of the sample, Pandemic Risk still appears in the news, but its Coverage is much closer to the levels of the other major risk themes.

Climate Risk receives a more persistent level of attention throughout the sample. Coverage is already relatively high in the early years and generally increases over time. Several periods stand out, including an increase around 2017, another around 2019–2020, and a stronger rise during 2021. Coverage reaches some of its highest levels in 2021 and again in 2023. Unlike Pandemic Risk, the evolution of Climate Risk is therefore not dominated by a single exceptional episode. Its presence in the news grows through several different periods.

Supply Chain Risk is initially less prominent, but its Coverage starts to increase towards the end of the 2010s. A clear rise appears in 2020, followed by another increase during 2021. Coverage remains relatively high through much of 2022 before declining during 2023. The series therefore shows a transition from a relatively limited presence in the earlier years to a much more visible role in the news after 2020.

AI-Related Risk remains the least covered theme for most of the sample. Its share of monthly news is very small during the earlier years and increases only gradually. The clearest change occurs in 2023, when AI Coverage rises noticeably and reaches its highest levels in the series. This is particularly interesting when compared with the Novelty results. The strong increase in AI-related attention during 2023 was not accompanied by an equally clear increase in average Novelty. The two indicators are therefore describing different changes in the information environment.

Cyber Risk receives a relatively stable but smaller share of news than Climate Risk and, after 2020, Pandemic and Supply Chain Risk. Its Coverage usually remains around 1–2%, although several clear peaks appear over time. Some of the largest increases occur around 2015, 2020, 2021, and 2022. The highest monthly Coverage is observed in early 2022, when the share of Cyber-related news rises above 3%.

For most of the sample, Cyber Risk accounts for roughly 1–2% of monthly news. This gives the series a relatively stable baseline, although it is interrupted by several short periods of higher attention. An early peak appears around the end of 2014, when Coverage rises well above its usual level. Other increases can be seen during 2016 and 2017, but they are generally temporary.

Cyber Coverage becomes somewhat stronger from 2019 onwards. The series often remains close to or above 2% during this period, and another clear peak appears at the beginning of 2020. Attention then falls sharply before recovering during the rest of the year.

Further increases appear during 2021. The most pronounced change, however, comes at the beginning of 2022. Monthly Coverage rises above 3%, reaching the highest level observed for Cyber Risk in the sample. The increase is short-lived, but it clearly separates this period from the usual level of Cyber-related attention.

After this peak, Coverage gradually returns to lower levels. During 2023, Cyber Risk generally accounts for a smaller share of monthly news, and Coverage falls particularly strongly towards the end of the year. This creates an interesting contrast with the Novelty results. The final months of 2023 were

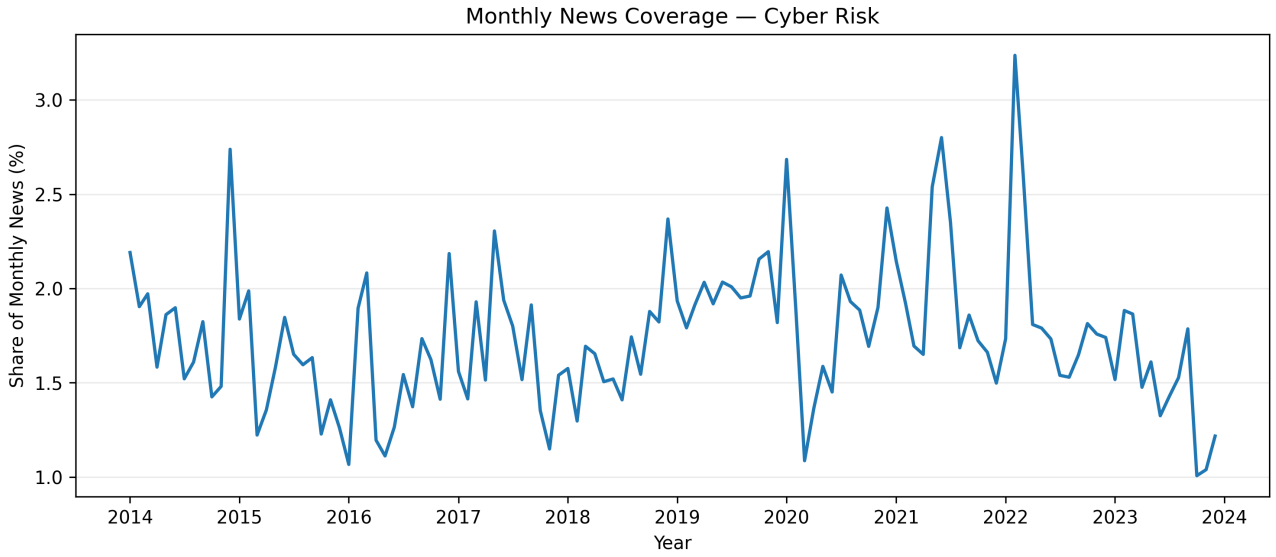


Figure 7.3: Monthly News Coverage for Cyber Risk, 2014–2023

characterised by the highest levels of Cyber Novelty, but not by particularly high News Coverage. In fact, Cyber Coverage was moving in the opposite direction.

The two measures therefore tell different parts of the story. At the beginning of 2022, Cyber Risk receives an unusually high level of news attention. This period is also close to the strongest Semantic Drift identified for Cyber Risk, with the maximum Drift occurring in February 2022. Towards the end of 2023, the situation is different. Cyber-related articles become more novel, but the theme does not receive more attention in the overall news flow. New information can therefore appear even when the risk is not becoming more prominent in the news.

7.2 Novelty

The temporal dimension is important because changes in Novelty identify periods in which the articles associated with a risk theme become more or less similar to the news published in the preceding year. Since the historical reference set is the broader news corpus rather than the risk-specific articles alone, Novelty measures how distinctive these articles are relative to the general information environment, not relative to earlier coverage of the same risk.

Figure 7.4 shows monthly mean Novelty separately for Cyber Risk, Climate Risk, Pandemic Risk, Supply Chain Risk, and AI-Related Risk.

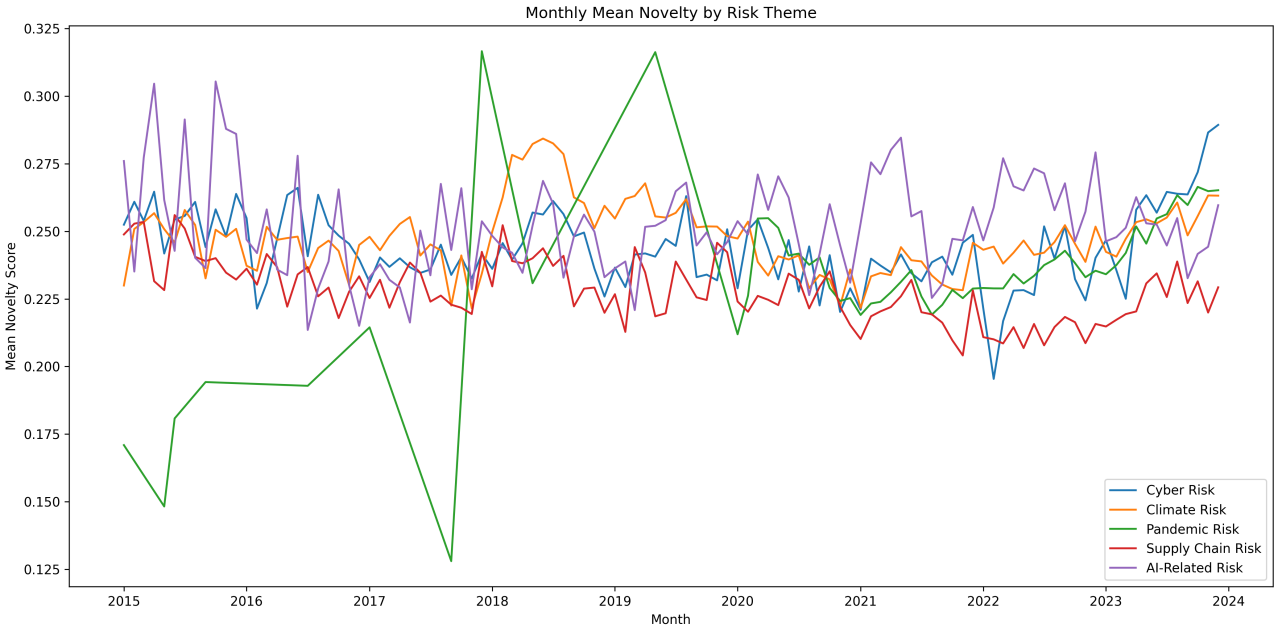


Figure 7.4: Monthly Novelty by Risk Theme, 2015–2023

The patterns for the five identified risk themes indeed vary.

With regards to **Climate Risk**, it is clear that there is an increase in Novelty starting from the year 2018 onwards. There is an upward trend in the initial period of the year, and the values stay at higher levels for several periods after this. Later on, Novelty starts to decline, with lower values being obtained at 2020 and 2021, and an increase is then found towards the end of the sample. The year 2018 is therefore the one where information regarding climate risk is highly novel compared to other periods in its time series.

As far as the **Supply Chain Risk** is concerned, Novelty starts at higher levels at the first years of the sample period, and decreases gradually in later periods. Lower values can be found around 2021 and 2022. It is then increasing again in 2023. It follows that there is no particular period during which Supply Chain risks become highly novel.

Pandemic Risk needs to be read with more caution. Before 2020, there are very few Pandemic-related articles in the dataset. For this reason, the large movements visible in the first part of the series can be driven by only one or a few observations and should not be treated as strong evidence of a change in Novelty. The series becomes much more informative from 2020, when the number of Pandemic-related articles increases substantially. Novelty decreases after the beginning of this period and reaches lower levels around 2021. It then starts to rise again, with a clearer increase during 2023.

AI-Related Risk is quite volatile, especially in the first years of the sample. Here too, the early values need to be considered together with the relatively small number of AI-related articles available at the time. In the later years the series becomes more stable. Novelty is relatively high in some periods around 2021 and 2022, but there is no clear increase during 2023. This is interesting because the number of AI-related articles grows strongly in that year. More news about a risk, therefore, does not necessarily mean that the information itself is becoming more novel.

Cyber Risk also changes considerably over time. For most of the sample, monthly Novelty moves within a relatively stable range, although there are several short-term increases and decreases. A particularly low value is visible at the beginning of 2022. Novelty then recovers and starts to rise more clearly during 2023. By the end of the year, the monthly average reaches the highest values observed for Cyber Risk in the sample.

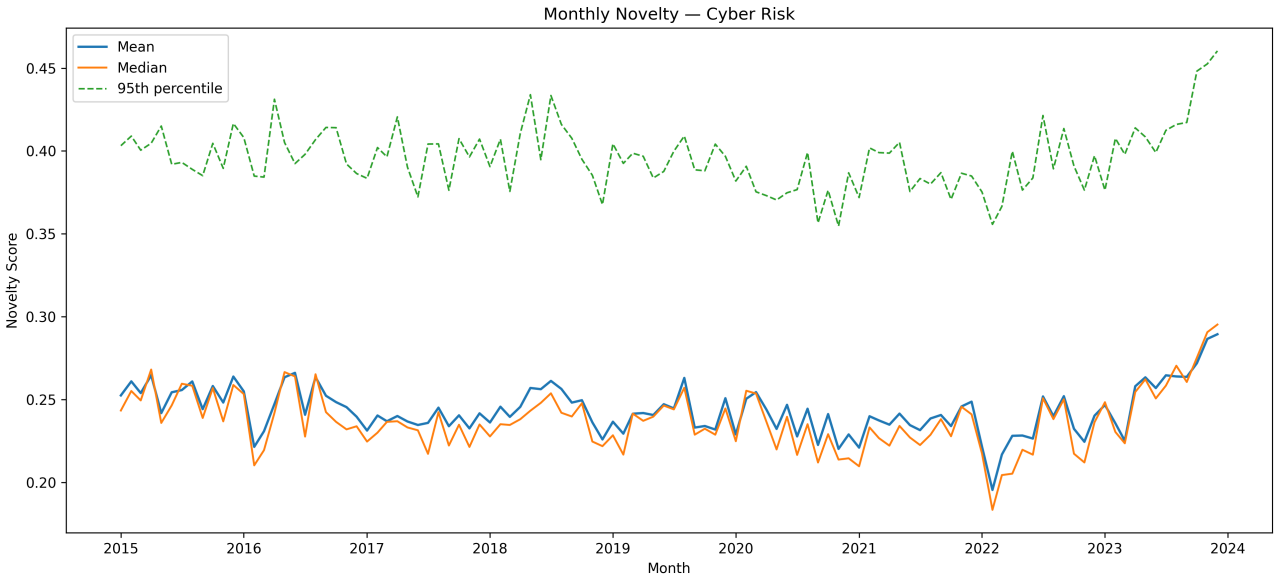


Figure 7.5: Monthly Novelty for Cyber Risk, 2015–2023

For most of the sample, the mean and median remain close to each other. Novelty is somewhat lower during parts of 2020 and 2021, and both measures fall sharply at the beginning of 2022. They then recover during the following months.

The most noticeable change comes in 2023. During the second half of the year, both the mean and the median start to increase. The rise becomes stronger in the final months, when they reach their highest levels in the series. Since the median increases together with the mean, this change does not appear to be driven only by a few highly novel articles. The typical Cyber-related article is also becoming more novel relative to the information that was available before it. The 95th percentile adds another part to the picture. It represents the upper end of the Novelty distribution and therefore captures the articles with particularly high Novelty scores. This measure also rises strongly at the end of 2023 and reaches its highest level in the sample. The increase is therefore visible both in the centre of the distribution and among the most novel Cyber-related articles. Taken together, these results make late 2023 the clearest period of high Novelty for Cyber Risk in the sample. Cyber-related articles become more different from the information that had recently been available, and the change is not limited to a small group of extreme observations. This does not mean that Cyber Risk itself emerged in late 2023. Novelty only captures one part of the process. What we can say from this indicator is that the information related to Cyber Risk became particularly novel during this period. The next indicators will help determine whether this change was also accompanied by changes in the content and attention associated with the risk.

A strong increase starting from February 2023 is evident. This could be related to the development of AI and, more generally, to the rapid growth of technology in recent years. This may have brought new topics and new risks in the cybersecurity field. For this reason, it would be interesting to

focus more on Cybersecurity Risk from 2023 onwards, to understand what is behind this increase. This will not be done in the current thesis, but it could be the next step of the research. In fact, to collect a new sample of news articles from 2023 onwards is needed, which is not available at the moment. This analysis will therefore be carried out in the future.

7.3 Semantic Drift

Figure 7.6 shows the monthly Semantic Drift for the five risk themes. Unlike Novelty, not every risk has an observation in every month. As explained in the previous chapter, Semantic Drift is only calculated when there is enough thematic information to construct a reliable monthly representation. Missing periods should therefore not be interpreted as zero drift.

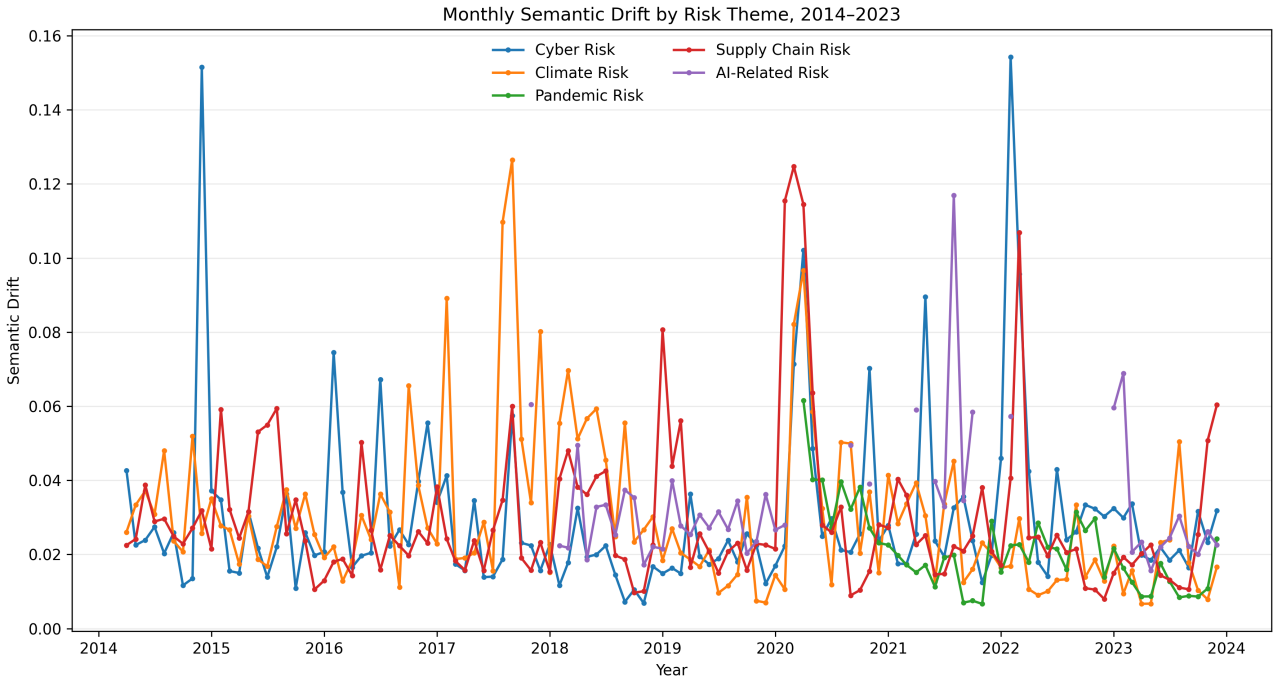


Figure 7.6: Monthly Semantic Drift by Risk Theme, 2014–2023

Most Semantic Drift values remain relatively low over the sample, but the series are interrupted by a number of clear peaks. These peaks are the most interesting part of the indicator. They identify months in which the content associated with a risk changes more strongly relative to its recent history.

Climate Risk shows several periods of higher Semantic Drift, especially between 2016 and 2018. The largest change appears around 2017, when Drift rises above 0.12. Other increases are visible before and after this point, but they are generally smaller. From 2019 onwards, the series is more stable, apart from a noticeable increase around the beginning of 2020. The graph therefore suggests that the content associated with Climate Risk changed more frequently during the earlier part of the sample and became relatively more stable in later years.

Supply Chain Risk also contains several clear changes. One appears around 2019, followed by a much stronger increase at the beginning of 2020. During this period, Semantic Drift rises above 0.12 before falling back towards its usual range. Another large increase is visible in early 2022. These

movements suggest that the meaning and content associated with Supply Chain Risk did not change gradually throughout the sample. Instead, the series is mostly stable and is interrupted by specific periods of stronger semantic change.

Pandemic Risk is only available from 2020 onwards because earlier months do not contain enough thematic information to construct a reliable monthly representation. The first available observations show relatively high Drift, followed by a decline during 2020 and 2021. After this initial period, most values remain at lower levels, generally with only moderate monthly movements. This suggests that the largest semantic adjustment occurred when Pandemic Risk first became strongly represented in the news, while its content became more stable afterwards.

AI-Related Risk is also unavailable for several parts of the sample. Where the indicator can be calculated, most observations remain at relatively moderate levels. The main exception occurs around the second half of 2021, when Semantic Drift rises sharply and reaches a value above 0.11. Other smaller increases appear in 2022 and 2023. The series therefore points to a few specific periods of semantic change rather than a continuous shift in AI-related content.

Cyber Risk shows several isolated peaks across the full sample. Some are visible in 2014, 2016, 2020, and 2021, but the largest occurs at the beginning of 2022.

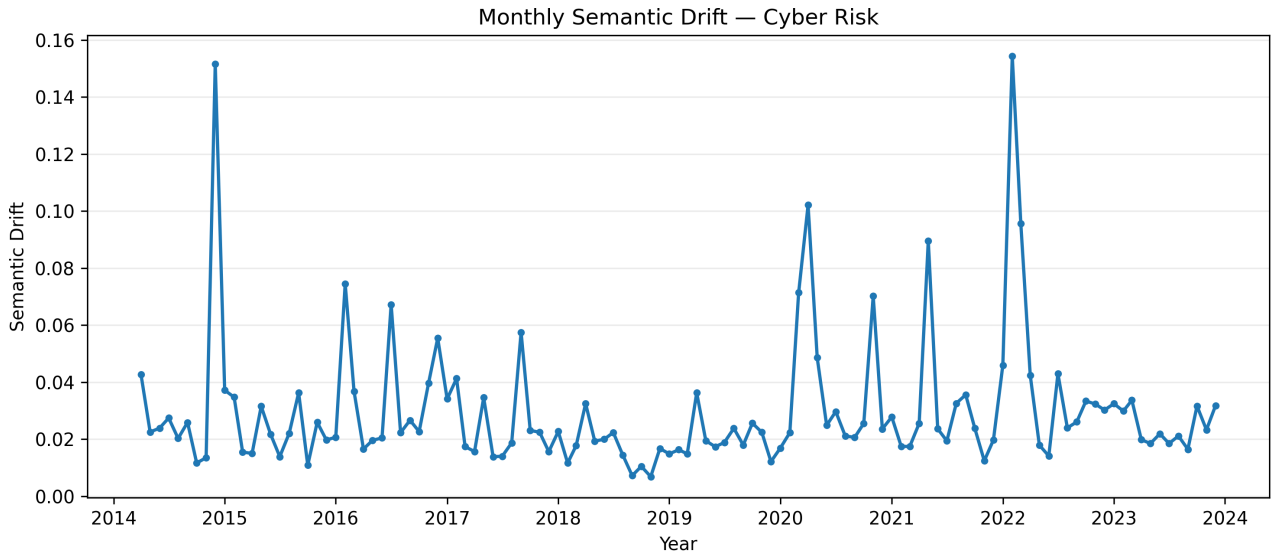


Figure 7.7: Monthly Semantic Drift for Cyber Risk, 2014–2023

For most of the sample, Cyber Semantic Drift remains relatively low. Values are often around 0.02–0.03, suggesting that the monthly semantic representation of Cyber Risk usually remains fairly close to its recent history. This does not mean that the content is unchanged. Rather, most monthly changes are small compared with the much larger movements that appear at specific points in the series.
The first major peak occurs towards the end of 2014, when Semantic Drift rises above 0.15. The increase is sharp and short-lived. Drift returns quickly to much lower levels in the following months. Several smaller peaks appear during 2016 and 2017, but none reaches the same magnitude.
A new period of stronger change appears in 2020. Semantic Drift rises above 0.10 before returning towards its usual range. Another noticeable increase occurs during 2021. These episodes show that the

semantic content of Cyber Risk changes at several points in the sample, rather than following a smooth long-term trend.

The largest value in the Cyber series occurs in February 2022, when Semantic Drift reaches approximately 0.154. This is a particularly clear departure from the surrounding months. The increase is followed by another relatively high observation and then by a rapid return towards lower levels. February 2022 therefore represents the strongest semantic change identified for Cyber Risk in the analytical period.

The rest of 2022 and 2023 looks quite different. Drift returns to much lower values and remains relatively stable, including during the final months of 2023. This is particularly interesting when compared with the Novelty results. Novelty increases strongly towards the end of 2023, but Semantic Drift does not show a similar increase. The two indicators are therefore capturing different aspects of the Cyber information environment.

The late-2023 Cyber articles become more novel relative to previously available information, but this does not appear to coincide with a major change in the overall semantic representation of Cyber Risk. By contrast, February 2022 is characterised by a strong change in the content associated with the theme. This difference is precisely why Novelty and Semantic Drift are treated as separate dimensions of risk emergence.

7.4 Burst Attention

While Coverage measures the amount of attention received by each risk, the Burst indicator identifies periods in which this attention increases unusually strongly relative to the risk's own recent history. A burst therefore does not simply correspond to high Coverage. It identifies an increase in attention that is unusually large compared with what was previously normal for that specific risk.

Figure 7.8 shows the evolution of Burst Intensity across the five risk themes. The results show different patterns. Some risks are characterized by one particularly strong episode, while others experience smaller but more recurrent bursts.

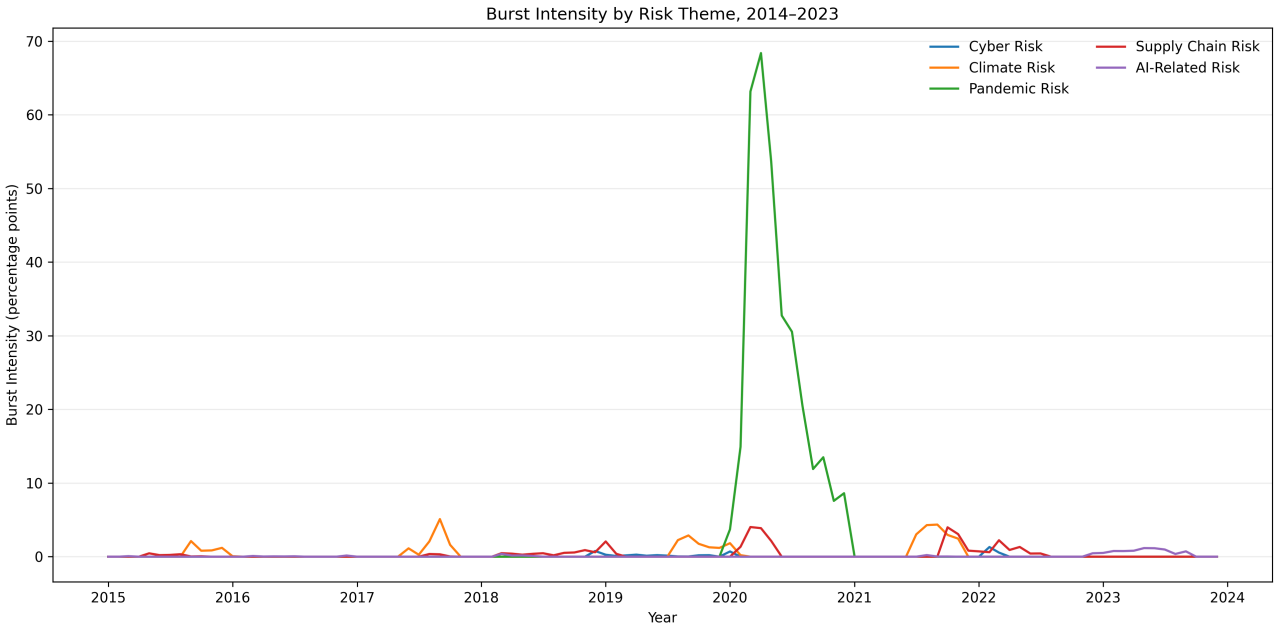


Figure 7.8: Burst Intensity by Risk Theme, 2014–2023

Pandemic Risk shows the most extreme burst in the sample. The episode begins in early 2020, when Burst Intensity increases very sharply and reaches almost 70 percentage points at its highest point. It then declines over the following months. This pattern is very different from the other risks. Pandemic Risk is not characterised by several separate bursts, but by one exceptionally large episode concentrated in 2020.

Climate Risk shows a more recurrent pattern. Several bursts appear over the sample, with episodes around 2015, 2017, 2019 and 2021. The strongest occurs in 2017, when Burst Intensity reaches slightly above 5 percentage points. These episodes are much smaller than the Pandemic burst, but they appear at different points in time. Climate Risk therefore experiences repeated periods in which attention increases unusually strongly relative to its own recent history.

Supply Chain Risk also experiences several burst episodes. They appear at different points in the sample and become particularly visible around 2018–2019, 2020 and 2021–2022. The strongest intensity is reached during the 2020 episode, at around 4 percentage points. Some of these episodes also continue for several months, suggesting that the increase in attention can remain unusually high for a relatively extended period.

AI-Related Risk shows bursts that are much smaller in absolute terms. This is consistent with the low level of AI Coverage observed for most of the sample. However, the indicator still detects several separate episodes. The clearest and most persistent one appears between the end of 2022 and 2023, at the same time as the increase in AI Coverage observed in the previous section. The Burst indicator therefore shows that this increase was also unusual relative to the previous history of AI-related attention.

Cyber Risk shows three distinct burst episodes. Their intensity is relatively small compared with the largest episodes observed for Pandemic, Climate and Supply Chain Risk, but they identify specific periods in which Cyber Coverage increased unusually strongly relative to its recent history. The first and longest episode occurs between the end of 2018 and 2019, followed by a short episode at the beginning of 2020. A final burst appears at the beginning of 2022.

Figure 7.9 provides a closer view of these episodes by showing the Cyber Coverage series together with the months classified as part of an active burst.

The first Cyber burst begins in December 2018 and continues until November 2019. It is the longest of the three detected episodes, lasting twelve months. During this period, Coverage remains unusually high relative to its previous history, although it does not increase continuously. This is important for understanding how the indicator works. Once a burst starts, it remains active while Coverage stays above its recent historical mean. A burst can therefore describe a period of sustained abnormal attention, rather than only a single monthly peak.

The second episode occurs in January 2020 and lasts only one month. Coverage reaches approximately 2.7%, but the increase is not sustained in the following months. The episode is therefore much shorter than the one observed in 2019.

The third burst occurs in February and March 2022. Although it lasts only two months, it is the strongest Cyber episode in terms of intensity. Coverage reaches approximately 3.24% in February, which is also the highest monthly Coverage observed for Cyber Risk in the sample. Burst Intensity reaches about 1.31 percentage points.

The figure also shows why Burst and Coverage should not be interpreted in the same way. Cyber

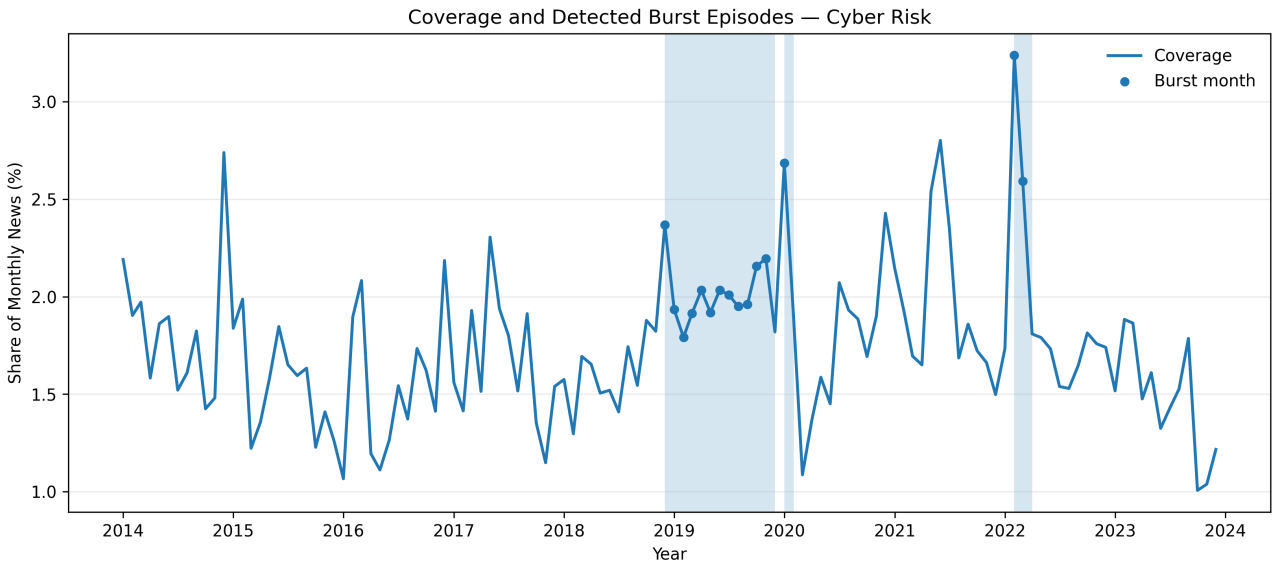


Figure 7.9: News Coverage and Detected Burst Episodes for Cyber Risk, 2014–2023

Coverage contains several visible peaks throughout the sample, but only some of them are classified as bursts. A high level of Coverage is not enough. The increase must be unusual relative to the recent history of Cyber Risk. The indicator therefore separates ordinary changes in attention from periods in which the increase is particularly strong compared with what came before.

The 2022 episode becomes especially interesting when the results are considered together with Semantic Drift. February 2022 is also the month in which Cyber Risk reaches its highest Semantic Drift, at approximately 0.154. The two indicators are measuring different things, but in this case they identify the same period. Attention to Cyber Risk was increasing unusually strongly, while the semantic content associated with the risk was also changing substantially relative to its recent history.

This differs from what is observed towards the end of 2023. The Novelty analysis showed a clear increase in the semantic novelty of Cyber-related articles during the final months of the sample. However, this period is not identified as a Cyber burst and Coverage is relatively low. The different indicators therefore do not always move together. In early 2022, the main signal comes from a change in attention and semantic content, while in late 2023 the strongest signal comes from the arrival of more novel information.

Chapter 8

Market Relevance of Emerging Risk Measures

The previous chapter examined Cyber Risk in details to prepare the financial analysis developed here. This chapter investigates whether news-based measures of cyber risk contain information about subsequent weekly equity portfolio excess returns. Coverage, Novelty, and Semantic Drift are examined through weekly regressions with six lags, following the empirical approach of Cherubini et al. Burst Detection is then used to examine periods of unusually intense media attention.
The analysis considers two equally weighted portfolios. The Diversified portfolio contains 195 firms from several industries. The Attacked + Security portfolio contains 76 firms: 66 companies classified as having experienced cyberattacks and 10 cybersecurity firms. The comparison allows the same news indicators to be evaluated for a broad collection of firms and for companies more directly connected to cybersecurity.
The empirical analysis begins with individual and combined tests of the news indicators. It then examines the contribution of Coverage to model fit and one-week-ahead forecasts. The final section studies whether the estimated relationships differ during Burst episodes and whether the Coverage results persist when contemporaneous Burst weeks are excluded.

8.1 Empirical Methodology

8.1.1 VAR specification

Let $R_{P,t}$ denote the weekly excess return of portfolio P in week t. The financial variable enters the regressions in levels. For each news indicator $I_t \in \{Coverage_t, Novelty_t, Drift_t\}$, the weekly change is defined as

$$\Delta I_t = I_t - I_{t-1}. \tag{8.1}$$

Weekly portfolio returns are already in terms of value changes and thus come into the analysis in level terms. The news indicators are in first difference terms to account for weekly changes in reporting intensity and semantics.
The lag order is set to six weeks based on the empirical framework used by Cherubini et al., 2024. This allows the analysis to pick up on the delayed relationship between cyber-risk news changes and portfolio returns within a six-week period. Using the same lag order for all indicators and portfolios ensures that statistical significance is compared on a similar framework.

For each portfolio and news indicator, the baseline bivariate VAR(6) is

$$\mathbf{Y}_{P,t} = \mathbf{c}_P + \sum_{\ell=1}^{6} \mathbf{A}_{P,\ell}\mathbf{Y}_{P,t-\ell} + \mathbf{u}_{P,t}, \qquad \mathbf{Y}_{P,t} = \begin{bmatrix} R_{P,t} \\ \Delta I_t \end{bmatrix}. \tag{8.2}$$

The financial equation used to assess the news indicator is

$$R_{P,t} = a_P + \sum_{\ell=1}^{6} b_{P,\ell} R_{P,t-\ell} + \sum_{\ell=1}^{6} c_{P,\ell}\Delta I_{t-\ell} + \varepsilon_{P,t}. \tag{8.3}$$

Each $c_{P,\ell}$ measures the conditional association between a change in the news indicator ℓ weeks earlier and the current weekly portfolio excess return. The coefficients are regression parameters from the return equation; they are not impulse responses to structural news shocks.

8.1.2 Tests and forecasting design

The null hypothesis for an individual indicator is

$$H_0 : c_{P,1} = \cdots = c_{P,6} = 0. \tag{8.4}$$

Joint restrictions are assessed with Wald tests based on heteroskedasticity and autocorrelation consistent (HAC) covariance estimates with a six-week bandwidth. The same procedure is applied to the relevant sets of lag coefficients in models containing multiple indicators. HAC(12) estimates provide a bandwidth sensitivity check for the individual and conditional results. Coefficients are estimated by ordinary least squares; HAC estimation affects their reported uncertainty.

Rejection of a joint restriction indicates that lagged changes in the relevant indicator are associated with weekly excess returns after controlling for lagged returns and any other included indicators. These tests concern the conditional relationship in the estimation sample. A separate expanding-window exercise evaluates one-week-ahead forecasts from an AR(6) benchmark and an augmented model containing six lags of $\Delta Coverage$. Forecast errors are compared using RMSE and out-of-sample R^2. The Clark–West test assesses the adjusted forecast-loss differential for the nested models, using one-sided HAC(6) inference.

8.2 Data and Portfolio Construction

8.2.1 News indicators and financial series

The three cyber-risk indicators are aggregated at weekly frequency using a Thursday-to-Wednesday calendar, with each observation assigned to the Wednesday marking the end of the corresponding week. Weekly Coverage is calculated as the proportion of cyber-risk-related articles among all news articles published during the week:

$$\text{Coverage}_t = \frac{N_t^{\text{cyber}}}{N_t^{\text{total}}}, \tag{8.5}$$

where N_t^{cyber} denotes the number of articles classified as cyber-risk-related and N_t^{total} is the total number of articles observed in week t.
Weekly Novelty is obtained by averaging the article-level novelty scores across cyber-risk-related articles published during the same week:

$$\text{Novelty}_t = \frac{1}{N_t^{\text{cyber}}} \sum_{i=1}^{N_t^{\text{cyber}}} \text{Novelty}_{i,t}, \tag{8.6}$$

Novelty measures the semantic distance between cyber-risk-related articles and their ten nearest historical semantic neighbours within the preceding 365 days. The historical reference set includes articles from the broader news corpus, rather than being restricted to previously identified cyber-risk-related articles.
Weekly Semantic Drift measures the distance between the semantic representation of current cyber-risk news and a historical reference constructed from previous weeks. The measure is based on cosine distance:

$$\text{Semantic Drift}_t = \frac{1}{B} \sum_{b=1}^{B} \left(1 - \frac{\mathbf{c}_t^{(b)\top} \mathbf{h}_t}{\|\mathbf{c}_t^{(b)}\| \|\mathbf{h}_t\|} \right), \qquad B = 200, \tag{8.7}$$

where $\mathbf{c}_t$ denotes the current weekly semantic centroid and $\mathbf{h}_t$ represents the historical reference. The reference is constructed using eligible weekly centroids from the preceding 52 weeks, requiring at least 13 eligible historical observations.
Semantic Drift is computed only for weeks containing at least 40 cyber-risk-related articles. This minimum-support requirement is intended to reduce the instability of semantic centroids estimated from small article samples, consistent with the reliability considerations discussed in Section 6.3. To limit the influence of differences in weekly news volume, the measure is calculated using a fixed sample size of 40 articles. For weeks containing more than 40 eligible articles, the final value is obtained by averaging the cosine distances computed over 200 random subsamples. Weeks with insufficient article support are treated as missing observations rather than assigned a zero value.

The financial regressions use the first weekly differences of the three indicators, defined as $\Delta I_t = I_t - I_{t-1}$, while portfolio excess returns enter in levels.

The Diversified portfolio consists of 195 companies from industries such as information technology, banks and pharmaceuticals. The Attacked + Security portfolio comprises 66 Attacked firms and 10 Security firms. These firms follow the classification established by Cherubini et al. Weekly average of all stock returns that are available for the specific week is calculated; then, the approximation of risk-free rate is subtracted and weekly excess returns are used as $R_{P,t}$ in financial equations.

Regression analysis is performed on a common sample of 376 usable weekly observations covering the period from 25 February 2015 to 5 April 2023. This sample is used for the individual, joint, and conditional tests of Coverage, Novelty, and Semantic Drift. An observation is included in the sample if portfolio excess returns and the three differenced news indicators, as well as their six weekly lags, are available.
The use of a common-sample restriction ensures that the indicators are tested using the same set of observations. Missing values in Semantic Drift, which result partly from its minimum article-count

requirement and historical reference requirements, reduce the number of usable weeks. All lagged variables are constructed on the original weekly calendar before observations with missing values are excluded. Consequently, the 376 observations in the common sample are not necessarily consecutive. The forecasting exercise considers only the Coverage indicator and therefore does not require the availability of Novelty and Semantic Drift. The dedicated forecasting sample contains 462 consecutive weekly observations covering the period from 25 February 2015 to 27 December 2023. Of these, 234 weeks are used for the initial estimation window, while the remaining 228 weeks are used to generate expanding-window one-week-ahead forecasts, from 21 August 2019 to 27 December 2023.

8.3 Financial Relevance of News Indicators

8.3.1 Individual indicators

Table 8.1 reports conventional joint tests and HAC(6) tests for the six lagged coefficients of each indicator. All regressions use the same 376 observations.

Table 8.1: Individual News Indicators: Joint Tests of Six Weekly Lags

	Diversified		Attacked + Security	
Indicator	Classical	HAC(6)	Classical	HAC(6)
Coverage	0.0036	**0.0049**	0.0008	**0.0009**
Novelty	0.7037	0.6394	0.3582	0.2381
Semantic Drift	0.9343	0.7488	0.8034	0.4457

Notes: Entries are p-values for the joint restriction that all six lagged coefficients of the named indicator equal zero in the $R_{P,t}$ equation. Classical refers to the conventional joint F-test. HAC(6) uses a six-week covariance bandwidth. Bold denotes significance at the 5% level under HAC(6).

Coverage is significant at the 1% level in both portfolios under HAC(6). Novelty and Semantic Drift have p-values above 0.05 in both cases. Coverage also remains significant when the HAC bandwidth is extended to twelve weeks: the corresponding p-values are 0.0010 for Diversified and 0.0001 for Attacked + Security. Neither Novelty nor Semantic Drift reaches the 5% threshold under HAC(12) (Novelty: 0.5476 and 0.1463; Semantic Drift: 0.6697 and 0.3339, for Diversified and Attacked + Security, respectively).

The evidence therefore points to Coverage as the strongest individual predictor among the three measures in these weekly return equations.

8.3.2 Combined indicators

Table 8.2 extends the tests to models containing two or three news indicators. Each indicator enters with six weekly lags, and the joint restriction covers all included news-indicator coefficients. The table reports HAC(6) inference.

Coverage and Semantic Drift are jointly significant in both portfolios. Coverage and Novelty are jointly significant in Attacked + Security, where the specification containing all three indicators is also significant. The Novelty–Semantic Drift combination is not significant in either portfolio. The combined tests impose more restrictions as indicators are added. For Diversified, Coverage is significant individually ($p = 0.0049$), while the Coverage–Novelty combination has $p = 0.0690$.

Table 8.2: Combined News Indicators: HAC(6) Joint Tests

Model	Diversified	Attacked + Security
Coverage + Novelty	0.0690	**0.0100**
Coverage + Semantic Drift	**0.0366**	**0.0077**
Novelty + Semantic Drift	0.8583	0.3568
Coverage + Novelty + Semantic Drift	0.1682	**0.0228**

Notes: Entries are HAC(6) joint-test p-values for all news-indicator lag coefficients in the $R_{P,t}$ equation. Each included indicator contributes six lags. All specifications use 376 observations. Bold denotes significance at the 5% level.

Adding coefficients with little detectable incremental contribution can reduce the power of a joint test. The larger joint p-value therefore does not overturn the individual Coverage result. The conditional tests below examine whether each indicator contributes separately when the others are included.

Does Coverage retain its contribution when other indicators are included?

The full three-indicator model is jointly significant for Attacked + Security ($p = 0.0228$). Table 8.3 tests each indicator's six lag coefficients while retaining the other indicators and six lags of weekly excess returns.

Table 8.3: Conditional Tests in the Full Three-Indicator VAR(6)

Restriction tested	Diversified	Attacked + Security
Coverage lags = 0, given Novelty and Drift	**0.0224**	**0.0078**
Novelty lags = 0, given Coverage and Drift	0.9893	0.7713
Drift lags = 0, given Coverage and Novelty	0.7906	0.7396
Novelty and Drift lags = 0, given Coverage	0.9813	0.7528

Notes: Entries are HAC(6) joint-test p-values. All regressions include six lags of $R_{P,t}$ and the six lags of each news indicator. The final row jointly tests twelve coefficients. Each regression uses 376 observations. Bold denotes significance at the 5% level.

Coverage remains significant after controlling for Novelty and Semantic Drift, with p-values of 0.0224 for Diversified and 0.0078 for Attacked + Security. The joint test of Novelty and Semantic Drift, conditional on Coverage, yields 0.9813 and 0.7528. In the Attacked + Security portfolio, the significant full-model test is accompanied by a significant conditional contribution from Coverage.

With HAC(12), Coverage remains significant ($p = 0.0241$ and $p = 0.0161$). The joint p-values for Novelty and Semantic Drift, conditional on Coverage, are 0.9751 and 0.6196.

Table 8.4 examines the reduced model containing Coverage and Semantic Drift. Each indicator enters with six weekly lags, alongside six lags of portfolio excess returns.

Coverage remains significant under HAC(6), with p-values of 0.0106 and 0.0029. Semantic Drift has conditional p-values of 0.8092 and 0.7409. With HAC(12), Coverage has p-values of 0.0057 and 0.0029, compared with 0.8080 and 0.6563 for Semantic Drift. The following section examines the lag coefficients of Coverage and its contribution to model fit.

Table 8.4: Conditional Tests in the Coverage–Semantic Drift VAR(6)

	Diversified		Attacked + Security	
Restriction tested	Classical	HAC(6)	Classical	HAC(6)
Coverage lags = 0, given Drift	0.0021	**0.0106**	0.0007	**0.0029**
Drift lags = 0, given Coverage	0.7441	0.8092	0.7290	0.7409

Notes: Entries are joint-test p-values for the six lags of the named indicator, conditional on the other news indicator and six lags of $R_{P,t}$. Each regression uses 376 observations. Bold denotes significance at the 5% level under HAC(6).

8.4 Financial Information Content of Coverage

Table 8.5 presents the estimated Coverage coefficients in the individual bivariate return equations. Individual p-values use HAC(6) standard errors.

Table 8.5: Coverage Coefficients by Weekly Lag: HAC(6) Inference

	Diversified		Attacked + Security	
Lag	Coefficient	p-value	Coefficient	p-value
1	0.8097	**0.0388**	0.9387	**0.0166**
2	0.4535	0.2007	0.6040	0.0906
3	1.1959	**0.0059**	1.4064	**0.0026**
4	0.5567	0.0544	0.5661	**0.0427**
5	0.3089	0.2862	0.2664	0.3820
6	0.2448	0.3726	0.1684	0.5771

Notes: Coefficients are estimated by OLS in the $R_{P,t}$ equation; p-values use HAC(6) standard errors. The regressors are weekly changes in Coverage at the indicated lag. Bold denotes significance at the 5% level. Coefficients are conditional regression parameters.

All six estimated coefficients are positive in both portfolios. The first and third lags are significant in each portfolio, while the fourth lag is also significant for Attacked + Security. The largest coefficients occur at lag three, reaching 1.1959 for Diversified and 1.4064 for Attacked + Security.

A positive coefficient indicates that a rise in Coverage relative to the preceding week is associated with a higher subsequent weekly excess return, conditional on the included lags. The estimated scale depends on the units of Coverage. The regressions do not identify the market mechanism behind this association, and the lag coefficients do not trace the full dynamic response to a structural news shock.

8.4.1 How much explanatory information does Coverage add?

To measure the contribution of Coverage, a restricted AR(6) return equation is compared with an augmented equation containing six additional lags of $\Delta Coverage$. Both models are fitted to the same 376 observations. The partial R^2 is

$$R^2_{\text{partial},P} = \frac{SSR_{\text{restricted},P} - SSR_{\text{Coverage},P}}{SSR_{\text{restricted},P}}, \tag{8.8}$$

where SSR denotes the residual sum of squares.

Adding Coverage raises adjusted R^2 by 3.61 percentage points for Diversified and 4.45 percentage points for Attacked + Security. The partial R^2 values indicate that the six Coverage lags remove 5.16%

Table 8.6: Incremental In-Sample Explanatory Contribution of Coverage

Measure	Diversified	Attacked + Security
R^2, financial lags only	0.0113	0.0385
R^2, with Coverage	0.0623	0.0972
Increase in R^2 (percentage points)	5.10	5.87
Adjusted R^2, financial lags only	-0.0048	0.0228
Adjusted R^2, with Coverage	0.0313	0.0673
Increase in adjusted R^2 (percentage points)	3.61	4.45
Partial R^2 of Coverage (%)	5.16	6.10
RMSE, financial lags only	0.02583	0.02777
RMSE, with Coverage	0.02515	0.02691
Reduction in RMSE (%)	2.61	3.10

Notes: Both equations contain an intercept and use 376 weekly observations. The restricted equation includes six lags of $R_{P,t}$; the augmented equation adds six lags of $\Delta Coverage$. RMSE is the square root of the mean squared fitted residual. Percentage-point changes in R^2 and percentage changes in RMSE are reported separately.

and 6.10% of the residual sum of squares remaining in the restricted financial equations. In-sample RMSE declines by 2.61% and 3.10%, respectively.

The relatively low R^2 of the restricted AR(6) equations reflects the limited linear dependence in weekly excess returns. The additional fit obtained with Coverage is measured against this benchmark on the same observations.

8.4.2 Financial interpretation of Coverage

Coverage measures the intensity of reporting about cyber risk. Its lagged weekly changes are positively associated with subsequent portfolio excess returns in both sets of firms. The association persists after the autoregressive return terms are included and after Semantic Drift is added to the equation.

The positive sign is consistent with several possible channels. Greater reporting may coincide with the diffusion of information about firms' ability to manage cyber incidents, or with broader developments that affect both news attention and equity markets. The available regressions do not distinguish these mechanisms. The observed coefficient pattern describes a conditional predictive association.

The fit comparison provides a measure of the association within the estimation sample. Forecast performance is examined separately because the regression coefficients must be re-estimated using information available before each forecasted week.

8.4.3 Out-of-Sample Forecasting Performance

The forecasting exercise compares an autoregressive benchmark with a model augmented by lagged Coverage. The benchmark is

$$R_{P,t} = \alpha_P + \sum_{k=1}^{6} \phi_{P,k} R_{P,t-k} + \varepsilon_{P,t}, \tag{8.9}$$

and the augmented model is

$$R_{P,t} = \alpha_P + \sum_{k=1}^{6} \phi_{P,k} R_{P,t-k} + \sum_{k=1}^{6} \beta_{P,k} \Delta Coverage_{t-k} + \varepsilon_{P,t}. \tag{8.10}$$

The dedicated Coverage sample contains 462 consecutive weekly observations. Both models are first estimated on 234 weeks. The estimation window then expands one week at a time, with coefficients fitted exclusively to observations preceding the forecasted week. This produces 228 one-week-ahead forecasts per portfolio from 21 August 2019 to 27 December 2023. No full-sample standardization of Coverage is performed within the forecasting procedure.

Forecast performance is summarized by RMSE and

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{t \in \mathcal{T}_{\text{OOS}}} (R_{P,t} - \widehat{R}^{Coverage}_{P,t})^2}{\sum_{t \in \mathcal{T}_{\text{OOS}}} (R_{P,t} - \widehat{R}^{AR}_{P,t})^2}. \tag{8.11}$$

A positive R^2_{OOS} indicates a lower mean squared forecast error for the augmented model over the evaluated weeks.

Table 8.7: Out-of-Sample Forecasting Performance of Coverage

Measure	Diversified	Attacked + Security
Out-of-sample observations	228	228
RMSE, AR(6)	0.036333	0.040462
RMSE, AR(6) + Coverage	0.036023	0.039976
RMSE reduction (%)	0.85	1.20
Out-of-sample R^2 (%)	1.70	2.39
Clark–West statistic, HAC(6)	1.9438	2.2530
Clark–West p-value	**0.0260**	**0.0121**

Notes: One-week-ahead expanding-window forecasts begin after 234 observations. The benchmark includes six lags of weekly excess returns; the augmented model also includes six lags of $\Delta Coverage$. Clark–West p-values are one-sided and use HAC(6) standard errors. Bold denotes significance at the 5% level.

The augmented model achieves a 0.85% lower RMSE for Diversified and a 1.20% lower RMSE for Attacked + Security. The corresponding out-of-sample R^2 values are 1.70% and 2.39%. Clark–West p-values of 0.0260 and 0.0121 indicate a significant adjusted forecast-loss improvement under the specified one-sided test. These p-values concern the Clark–West adjustment for nested models rather than a direct test of the raw RMSE difference.

The OOS RMSE exceeds the fitted residual RMSE reported in Table 8.6. The OOS figures measure errors on observations excluded from each estimation window and cover a different evaluation sample. To assess the sensitivity of the forecasting results to the initial estimation period, Table 8.8 reports the results obtained with four alternative initial window lengths.

The augmented model achieves lower RMSE in all four forecasting configurations. Clark–West p-values remain below 0.05 in both portfolios. Because the first forecast date changes with the initial estimation window, these comparisons measure sensitivity to the forecasting design across different evaluation periods. The overlapping forecast periods limit their interpretation as separate confirmations.

Table 8.8: Out-of-Sample Forecasting Robustness Across Initial Estimation Windows

		Diversified		Attacked + Security	
Initial window	OOS forecasts	RMSE gain (%)	CW p-value	RMSE gain (%)	CW p-value
156	306	0.62	0.0330	0.89	0.0195
208	254	0.89	0.0221	1.23	0.0106
234	228	0.85	0.0260	1.20	0.0121
260	202	0.81	0.0325	1.20	0.0137

Notes: RMSE gain is the percentage reduction relative to the AR(6) benchmark. CW p-values are one-sided Clark–West tests using HAC(6) standard errors. Each initial window produces a different out-of-sample evaluation period. The four evaluation samples overlap substantially and are not independent replications.

8.5 News-Based Indicators During Burst Episodes

The previous sections establish an association between lagged Coverage and subsequent weekly excess returns. This section examines weeks when cyber-risk reporting increases unusually sharply relative to its recent history.

The monthly Burst Detection procedure introduced in Section 6.4 is adapted to weekly observations. For each week, Coverage is compared with the mean and sample standard deviation of the preceding 52 weeks, excluding the current week. The standardized shock is

$$x_{N,t}^{(52)} = \frac{k_{N,t} - v_{N,t}^{(52)}}{w_{N,t}^{(52)}}, \tag{8.12}$$

where $k_{N,t}$ is normalized weekly Coverage and $v_{N,t}^{(52)}$ and $w_{N,t}^{(52)}$ are its trailing mean and sample standard deviation. The sample standard deviation is calculated with denominator $n - 1$, where n is the number of observations in the trailing window.

A Burst begins when the standardized shock exceeds three and current Coverage exceeds the 95th percentile of preceding observations. Once activated, the episode remains active while Coverage exceeds its rolling historical mean. Burst State identifies active weeks; Burst Intensity measures excess Coverage above that mean.

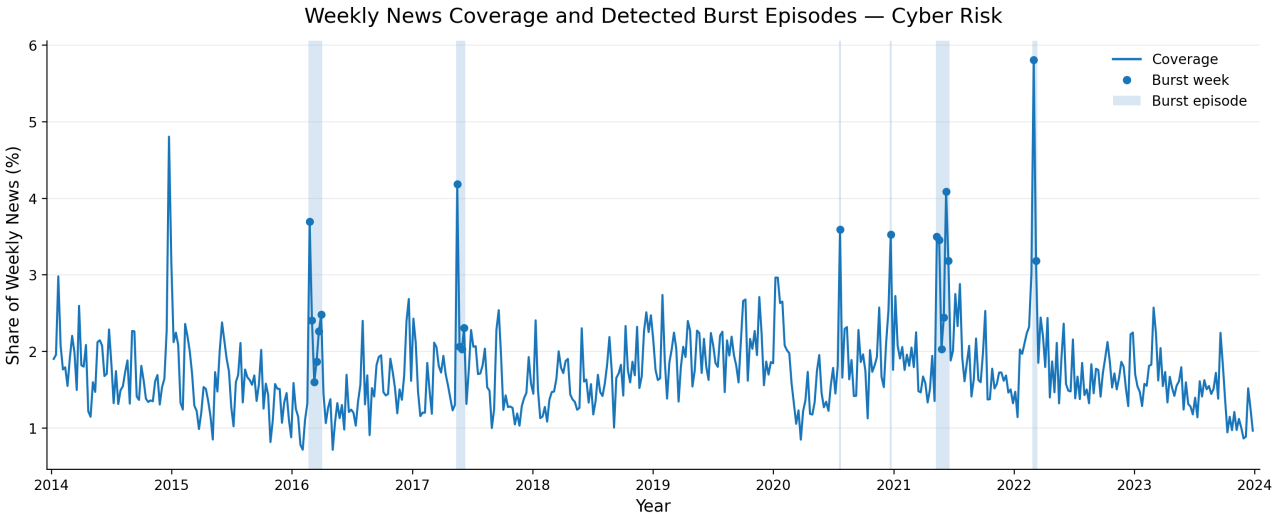


Figure 8.1: Weekly Cyber-Risk Coverage and detected Burst episodes, 2014–2023. The shaded intervals indicate weeks classified as Burst.

The first 52 weeks provide the historical reference window; elevated Coverage before that window is complete cannot trigger a Burst classification. The weekly procedure identifies six Burst episodes over 2014–2023. Their starting months are February 2016, May 2017, July 2020, December 2020, May 2021, and February 2022. After alignment with the common financial sample, 19 of the 376 regression observations are classified as Burst weeks.

8.5.1 Information characteristics during Burst episodes

Table 8.9: News-Based Indicators During Ordinary and Burst Weeks

Indicator	Ordinary weeks	Burst weeks	Difference
Coverage	0.01718	0.02835	+65.1%
Novelty	0.24301	0.23075	-5.0%
Semantic Drift	0.05987	0.08276	+38.2%
Observations	357	19	

Notes: The table reports indicator levels averaged across ordinary and Burst weeks in the common regression sample. Differences are expressed relative to the ordinary-week average. No formal test of differences is reported.

Coverage is higher during Burst weeks by construction, since exceptional Coverage is used to identify the episodes. Novelty averages 0.23075 during Burst weeks, compared with 0.24301 in ordinary weeks. Semantic Drift averages 0.08276 during Burst weeks, compared with 0.05987 in ordinary weeks. These differences describe the 19 Burst observations; their statistical significance is not assessed here.

8.5.2 Does financial relevance change during Burst periods?

A separate VAR(6) estimated only on Burst weeks would have too few observations to estimate its lag coefficients reliably. The 19 contemporaneous Burst observations also occur in separated episodes. The analysis therefore retains the common sample and uses a parsimonious interaction specification.

For indicator I, define the average of its preceding six weekly changes as

$$S_{I,t} = \frac{1}{6} \sum_{\ell=1}^{6} \Delta I_{t-\ell}. \tag{8.13}$$

The return equation is

$$\begin{aligned} R_{P,t} = \alpha_P + \sum_{\ell=1}^{6} \beta_{P,\ell} R_{P,t-\ell} + \gamma_P S_{I,t} \\ + \theta_P B_{t-1} + \delta_P (S_{I,t} B_{t-1}) + \varepsilon_{P,t}, \end{aligned} \tag{8.14}$$

where B_{t-1} indicates whether the preceding week was classified as Burst. The interaction coefficient δ_P measures the difference in the conditional association across the two states. Each indicator and portfolio is estimated separately, and the test concerns

$$H_0 : \delta_P = 0. \tag{8.15}$$

Table 8.10: Burst-Regime Interactions: HAC(6) Tests

Indicator	Diversified	Attacked + Security
Coverage	0.2761	0.2483
Novelty	0.3242	0.4319
Semantic Drift	0.8639	0.8145

Notes: Entries are HAC(6) p-values for the interaction between the mean of six lagged weekly indicator changes and the preceding week's Burst State. Each regression uses 376 observations. The common sample contains 19 contemporaneous Burst weeks and 20 observations with lagged Burst State equal to one. The lagged count differs because Burst State is shifted on the original weekly calendar before the 376 valid regression dates are selected.

For Coverage, the main-effect coefficient γ_P has HAC(6) p-values of 0.1066 for Diversified and 0.0825 for Attacked + Security. This coefficient describes the association when the preceding week is classified as ordinary. None of the interaction coefficients is significant at the 5% level. The small number of Burst observations limits the precision of these estimates, so the tests cannot establish that the associations are identical across states. Within the fitted interaction specifications, the data do not identify a statistically significant difference.

8.5.3 Coverage significance after excluding Burst weeks

The individual Coverage regression is also re-estimated after removing observations whose contemporaneous week is classified as Burst. The lagged variables retain their original weekly alignment, and the same HAC(6) joint test is applied.

Coverage remains significant in both portfolios after the 19 contemporaneous Burst observations are excluded. This comparison shows that the full-sample result persists among the retained financial observations. Coverage recorded during Burst episodes can still enter regressions for later non-Burst weeks through the six lagged regressors, so the restricted sample retains some information originating in Burst periods.

Table 8.11: Coverage Significance After Excluding Burst Weeks

Sample	Observations	Diversified	Attacked + Security
Full sample	376	**0.0049**	**0.0009**
Excluding Burst weeks	357	**0.0041**	**0.0004**

Notes: Entries are HAC(6) p-values for the joint significance of six lagged $\Delta Coverage$ coefficients in the $R_{P,t}$ equation. The restricted sample excludes 19 observations classified as Burst in the contemporaneous week. Regressors are constructed using the original chronology. Bold denotes significance at the 5% level.

8.6 Comparison with Cherubini et al., 2024

The results obtained for Coverage can be compared with those reported by Cherubini et al. in Text-Based Measures of Cyber Risk: The Economic Impact of Cyber Risk News. Both analyses examine the relationship between media attention to Cyber Risk and subsequent equity returns using a VAR framework with six weekly lags. However, differences in the financial variable specification and portfolio construction need to be considered when comparing the estimated results.

Cherubini et al. find that increases in Cyber Risk news attention are followed by positive equity market responses, with the strongest effect occurring approximately four weeks after the initial shock. Although some sectors exhibit negative responses during the first week, the estimated effect becomes positive across all sectors at the four-week horizon. The positive effect is particularly significant for cybersecurity providers, suggesting that increased attention to Cyber Risk may also reflect opportunities associated with cybersecurity activities.

The present analysis provides evidence consistent with the positive relationship identified by Cherubini et al. Coverage is jointly significant in both portfolios, with HAC(6) p-values of 0.0049 for Diversified and 0.0009 for Attacked + Security. Furthermore, all six estimated Coverage coefficients are positive in both portfolios, indicating that increases in media attention are associated with higher subsequent weekly excess returns, conditional on past returns.

A difference emerges in the temporal distribution of the estimated coefficients. While Cherubini et al. identify the strongest positive effect at the fourth weekly lag, the largest Coverage coefficients in the present analysis occur at the third lag, reaching 1.1959 for Diversified and 1.4064 for Attacked + Security. The first and third lags are statistically significant in both portfolios, while the fourth lag is also significant for Attacked + Security. Nevertheless, the estimated dynamics are not directly comparable, as the two studies adopt different specifications of the financial variable. The results therefore indicate a broadly similar positive relationship over subsequent weeks, rather than an identical temporal pattern.

The comparison across portfolios provides further evidence of consistency with Cherubini et al. Their analysis identifies particularly strong effects among cybersecurity providers. In the present study, Coverage is statistically significant in both portfolios, but its explanatory contribution is larger for Attacked + Security. The partial R^2 associated with Coverage reaches 6.10% for Attacked + Security, compared with 5.16% for Diversified. This result suggests that Cyber Risk reporting may contain more relevant information for firms directly connected to cybersecurity. However, since Attacked + Security includes both previously attacked firms and cybersecurity providers, the contribution of these two groups cannot be separately identified.

Overall, the results support the main finding of Cherubini et al. that media attention to Cyber Risk is associated with subsequent equity market performance. Although the estimated temporal patterns

differ, both analyses identify a positive relationship over a horizon of several weeks. The present findings also indicate that Coverage contains statistically significant information about subsequent excess returns in both portfolios, with a somewhat stronger explanatory contribution for Attacked + Security.

8.7 To Sum Up

Coverage has the clearest statistical association with subsequent weekly portfolio excess returns among the three indicators examined. Its six lagged changes are jointly significant in both portfolios, after controlling for the other two indicators. Adding Coverage raises adjusted R^2 by 3.61 percentage points for Diversified and 4.45 percentage points for Attacked + Security.

The expanding-window exercise produces lower one-week-ahead forecast RMSE with Coverage in both portfolios. The reductions are 0.85% and 1.20%, with one-sided Clark–West HAC(6) p-values of 0.0260 and 0.0121. The alternative initial estimation windows also produce positive RMSE gains, although their evaluation periods overlap. The Burst interaction tests find no significant change in the estimated association across regimes, while the Coverage joint tests remain significant after contemporaneous Burst weeks are excluded.

These findings document a measurable predictive association in the specified weekly samples. The forecasting evidence concerns the portfolios and evaluation dates examined here.

Chapter 9

Conclusions

This thesis investigated how financial news can be used to measure the evolution of emerging risks and whether the resulting measures contain information relevant to financial markets. The analysis focused on five risk categories: Cyber Risk, Climate Risk, Pandemic Risk, Supply Chain Risk, and AI-Related Risk. The objective was to move beyond a simple count of news articles and examine different aspects of how risks appear and evolve in the information environment.

The research was developed in two main parts. The first constructed a framework for identifying risk-related articles and measuring changes in their frequency and semantic content. The second applied the resulting indicators to Cyber Risk and investigated their relationship with subsequent weekly equity portfolio excess returns. The empirical results show that the indicators capture different dimensions of risk-related reporting and that, among the measures examined in the financial analysis, Coverage provides the clearest evidence of predictive information.

The starting point was the construction of a consistent news database. The original material covered articles published between 2014 and 2023 by five newspapers: the Financial Times, Wall Street Journal, New York Times, Los Angeles Times, and USA Today. Before using these articles to measure risk-related attention, it was necessary to verify their metadata, standardize their structure, and address duplicated content. This step was important because repeated articles could artificially increase the measured frequency of a risk theme, while inconsistencies in publication dates or sources could affect comparisons over time.

The database initially contained 2,374,643 records in the staging table. After validation, 2,373,999 articles were transferred to the main database. The subsequent identification of 70,964 exact duplicates and 53,780 near-duplicates produced a final analytical corpus of 2,249,255 articles. The resulting dataset provided a common and consistent basis for the classification and measurement procedures developed in the following chapters.

The next task was to identify which articles were related to the five selected risks. This was not treated as a simple assignment of each article to one category. A news article can discuss more than one risk, and its relevance to each theme can vary. The classification framework therefore assigned continuous thematic relevance scores to article–risk pairs.

Three approaches were developed: keyword-based scoring, weakly-supervised scoring, and supervised scoring. The keyword approach provided a transparent lexical baseline, but its results depended on the presence of predefined expressions. Weak supervision introduced contextual labeling rules to capture more specific evidence of thematic relevance. The supervised approach used TF–IDF representations and logistic regression to learn associations between textual features and the risk categories.

The comparison of these approaches was important because the identification of relevant articles directly affects the subsequent indicators. An overly restrictive classification can exclude articles that discuss a risk without using its most common terminology. Conversely, an overly broad classification can include unrelated reporting and distort the measured evolution of the theme. The use of manually reviewed observations and classification diagnostics therefore served to evaluate the reliability of the thematic scores before their application to the complete corpus.

The final scoring procedure evaluated each of the 2,249,255 articles against all five risk categories. This produced 11,246,275 article–theme combinations for each methodology and 33,738,825 method–article–theme observations across the three approaches. These scores made it possible to measure thematic relevance systematically over a large collection of financial news.

Thematic classification alone, however, could not describe how the information contained in the articles changed over time. Two articles may concern the same risk while discussing substantially different events or developments. Similarly, articles may use different words to describe closely related information. For this reason, the thesis introduced document embeddings as a second representation of the corpus.

The embeddings mapped articles into a common semantic space, allowing their textual content to be compared through vector similarity. This representation complemented thematic relevance scoring. While classification identified the risk categories associated with each article, embeddings provided information about the similarity between articles and about changes in the content associated with a theme.

Combining these two representations made it possible to construct four indicators: Novelty, Semantic Drift, Coverage, and Burst Detection. Each was designed to capture a different characteristic of risk emergence.

Novelty measured how different incoming risk-related articles were from news published in the preceding year. The historical comparison was based on the broader news corpus rather than only on previous articles concerning the same risk. Semantic Drift measured changes in the semantic representation of a risk category over time. Coverage measured the share of news devoted to a given risk, while Burst Detection identified periods in which that attention increased unusually strongly relative to the theme's recent history.

The distinction between these measures was central to the empirical analysis. An increase in Coverage does not necessarily imply that the reported information is new. A risk may receive sustained attention because newspapers repeatedly discuss developments that are already familiar. Conversely, Novelty may increase when new information appears even if the overall volume of reporting remains limited. Semantic Drift adds another dimension by identifying changes in the content associated with a risk category, while Burst Detection distinguishes exceptional increases in attention from ordinary fluctuations in Coverage.

The analysis of the five risk categories showed that these distinctions are empirically relevant. The indicators did not follow a common pattern across themes, and their movements often differed even within the same category.

Climate Risk displayed relatively persistent news attention, with several periods of increasing Coverage rather than a single dominant episode. Its Novelty was particularly high during 2018, indicating that changes in the informational content of climate-related reporting did not simply coincide with its periods of greatest overall attention.

Supply Chain Risk became substantially more visible in the news after 2020, with elevated Coverage

during 2021 and much of 2022. Its Novelty, however, followed a more gradual pattern and remained relatively low around 2021–2022. This contrast illustrates how a theme can become increasingly prominent without a corresponding increase in the distinctiveness of incoming articles.

Pandemic Risk showed a different trajectory, characterized by a major change in its representation in the news beginning in 2020. The limited number of relevant articles before that period also demonstrated an important measurement issue: semantic indicators based on very few observations can be unstable and must be interpreted cautiously. Once Pandemic Risk became widely represented, the indicators provided a more informative description of how reporting evolved.

AI-Related Risk remained comparatively marginal in the news for much of the sample before experiencing a pronounced increase in Coverage during 2023. This increase was also reflected in the detection of unusual attention episodes. However, average Novelty did not show an equally clear rise during the same year. The result indicates that the rapid expansion of reporting about a risk does not necessarily correspond to an equivalent increase in semantic novelty.

These differences support the use of a multidimensional measurement framework. The emergence of a risk in the news cannot be described adequately by a single indicator because attention, informational novelty, and semantic change do not necessarily evolve together.

Cyber Risk was examined in greater detail, both because of its relevance to the financial application and because its indicators showed several distinct episodes over the sample period. Its Coverage generally remained around 1–2% of monthly news, although this relatively stable baseline was interrupted by periods of stronger attention.

The monthly Burst Detection procedure identified three main Cyber Risk episodes. The first extended from December 2018 to November 2019, representing a sustained period of unusually high attention. A shorter episode occurred in January 2020. The third took place in February and March 2022 and was the strongest in terms of intensity.

February 2022 was particularly important because different indicators pointed to substantial changes during the same period. Cyber Coverage reached approximately 3.24%, its highest monthly value in the sample, while Semantic Drift reached approximately 0.154. The coincidence of these movements indicates that unusually intense reporting was accompanied by a substantial change in the semantic content associated with Cyber Risk.

A different pattern emerged towards the end of 2023. Cyber Novelty increased markedly, with the mean, median, and upper part of its distribution all reaching high values. At the same time, Coverage was relatively low and the period was not classified as a Burst episode. This suggests that incoming Cyber-related information had become more distinct from news published during the preceding year without producing a comparable increase in the theme's overall share of news.

The comparison between these two periods provides one of the clearest findings of the descriptive analysis. In early 2022, Cyber Risk was characterized by an exceptional increase in attention accompanied by strong semantic change. In late 2023, the most prominent signal was the increased novelty of the information being published. The indicators therefore identified different forms of change within the same risk category.

The final part of the thesis investigated whether the constructed Cyber Risk indicators were associated with subsequent equity returns. Following the empirical approach developed in the literature on cyber-risk news and financial markets, the analysis considered weekly changes in Coverage, Novelty, and Semantic Drift, together with six weekly lags of portfolio excess returns.

Two equally weighted portfolios were examined. The Diversified portfolio contained 195 firms from

different industries, while the Attacked + Security portfolio contained 76 firms, including companies previously affected by cyberattacks and firms operating in cybersecurity. This distinction allowed the indicators to be evaluated for a broad group of companies and for firms more directly connected to cybersecurity.

The common regression sample contained 376 usable weekly observations, covering the period from February 2015 to April 2023. The same observations were used for the individual, combined, and conditional tests of the three indicators. The forecasting exercise considered Coverage separately and therefore used a longer sample extending to December 2023.

The empirical results showed a clear difference between the financial information contained in the three indicators. Coverage was the only measure whose six lagged changes were jointly significant in both portfolios when the indicators were examined individually. Under HAC(6) inference, the corresponding p-values were 0.0049 for the Diversified portfolio and 0.0009 for the Attacked + Security portfolio. Novelty and Semantic Drift did not reach the same significance threshold.

The importance of Coverage was also confirmed by the combined specifications. When all three indicators were included, the six Coverage lags remained jointly significant conditional on Novelty and Semantic Drift, with HAC(6) p-values of 0.0224 and 0.0078 for the two portfolios. By contrast, the additional contribution of Novelty and Semantic Drift was not statistically significant once Coverage was included.

These results indicate that the three indicators do not contribute equally to the prediction of weekly excess returns. Novelty and Semantic Drift remain useful for describing the informational evolution of Cyber Risk, but the financial regressions do not provide comparable evidence that they contain incremental return-predictive information over the sample examined.

The estimated Coverage coefficients were positive across all six lags in both portfolios. This means that increases in Cyber Risk Coverage were associated with higher subsequent weekly portfolio excess returns, conditional on the included lagged variables. The positive relationship should not, however, be interpreted as evidence that greater cyber-risk attention causes stock prices to rise. The regressions establish a conditional predictive association, but they do not identify the economic mechanism responsible for it.

One possible interpretation is that greater Cyber Risk Coverage also provides information about how firms respond to cyber incidents and their ability to maintain operations despite these events. In this case, the positive relationship could be consistent with information about corporate resilience being gradually reflected in market prices. Another possibility is that changes in news attention and subsequent returns are influenced by broader market conditions or developments affecting both variables. These explanations are consistent with the estimated results, but the analysis does not establish which mechanism is responsible for the observed relationship.

The contribution of Coverage was also assessed by comparing the explanatory power of an autoregressive return equation with that of a model augmented by six lags of changes in Coverage. Adding Coverage increased adjusted R^2 by 3.61 percentage points for the Diversified portfolio and 4.45 percentage points for the Attacked + Security portfolio. The corresponding partial R^2 values were 5.16% and 6.10

These in-sample results were complemented by an out-of-sample forecasting exercise. Using an expanding estimation window, the thesis compared one-week-ahead forecasts from an AR(6) benchmark with forecasts from the same model augmented by lagged Coverage. The evaluation included 228 forecasts for each portfolio, covering the period from August 2019 to December 2023.

The model including Coverage achieved a lower root mean squared forecast error in both cases. The

RMSE reduction was 0.85% for the Diversified portfolio and 1.20% for the Attacked + Security portfolio, corresponding to out-of-sample R^2 values of 1.70% and 2.39%. The one-sided Clark–West tests also indicated statistically significant improvements, with HAC(6) p-values of 0.0260 and 0.0121. The forecasting gains were relatively modest in magnitude, but they were observed in both portfolios and were supported by the adjusted forecast-comparison tests. The results were also consistent across alternative initial estimation-window lengths. This strengthens the evidence that Cyber Risk Coverage contains information not fully captured by the portfolios' own recent returns, within the forecasting periods and specifications considered.

The improvement in forecasting accuracy does not necessarily imply an equivalent economic benefit. Although the model including Coverage produced statistically supported reductions in forecast errors, the analysis did not examine whether these improvements could be translated into profitable investment decisions. Such an assessment would require an explicit trading strategy and the consideration of transaction costs, portfolio turnover, and risk-adjusted returns. The forecasting results therefore provide evidence of predictive information, but they do not establish the economic profitability of using Coverage in an investment strategy.

The final empirical tests examined whether the relationship between the news indicators and returns changed during periods of unusually intense Cyber Risk attention. For this purpose, the Burst Detection procedure was adapted to weekly data.

The weekly procedure identified six Burst episodes over 2014–2023. Within the common financial regression sample, 19 of the 376 observations were classified as contemporaneous Burst weeks. During these weeks, average Coverage was substantially higher than in ordinary weeks, as expected from the definition of the indicator. Average Semantic Drift was also higher, while average Novelty was slightly lower. These comparisons were descriptive and were not accompanied by formal tests of the differences.

Interaction regressions were then used to investigate whether the estimated relationships between lagged changes in the indicators and portfolio returns differed following Burst periods. The tests did not identify statistically significant differences between the two regimes for Coverage, Novelty, or Semantic Drift. Given the small number of Burst observations, this finding does not establish that the relationships are identical. It indicates only that the estimated specifications did not provide sufficient evidence of a difference.

A further sensitivity check excluded the 19 contemporaneous Burst weeks from the Coverage return regressions. Coverage remained jointly significant in both portfolios, with HAC(6) p-values of 0.0041 for Diversified and 0.0004 for Attacked + Security. The result suggests that the predictive association identified in the full sample was not driven exclusively by the observations classified as contemporaneous Burst weeks. Nevertheless, the retained regressions could still contain lagged information originating during earlier Burst periods.

Taken together, the empirical findings lead to two main conclusions.

First, financial news can be used to construct complementary measures of how emerging risks evolve. The analysis across five categories demonstrates that changes in attention, novelty, and semantic content are related but distinct phenomena. The Cyber Risk application illustrates this particularly clearly: the exceptional attention and semantic change observed in early 2022 differed from the increase in informational novelty observed at the end of 2023. A framework based only on article frequency would not distinguish these patterns. These findings concern the descriptive measurement of Cyber Risk and do not, by themselves, establish the financial relevance of the individual episodes.

Second, the financial relevance of these measures is not uniform. In the Cyber Risk application, Coverage provides the strongest and most consistent evidence of an association with subsequent weekly equity excess returns. Its contribution remains statistically significant when the other semantic indicators are included, improves in-sample model fit, and produces modest but statistically supported out-of-sample forecasting gains. Novelty and Semantic Drift contribute to the measurement of risk-related information, but the financial analysis does not establish an equivalent incremental predictive contribution for these measures.

Several limitations qualify these conclusions. The analysis is based on articles from five newspapers and therefore measures risks as represented in this particular information environment, rather than all information available to market participants. The classification and semantic indicators depend on the selected methodologies and their implementation choices. In particular, Novelty is measured relative to the broader news corpus of the preceding year rather than relative to earlier coverage of the same risk. Its interpretation can therefore differ from that of a measure based exclusively on previous articles about a particular risk category. The reliability of semantic indicators can also be affected when a risk category is represented by relatively few articles.

The financial results are specific to Cyber Risk, the two constructed portfolios, the weekly sample, and the regression and forecasting specifications used in the thesis. The common regression sample ends in April 2023, so the increase in Cyber Novelty observed towards the end of 2023 is not covered by the financial tests involving all three indicators. Although this episode is relevant to the descriptive analysis, the results do not establish whether the corresponding increase in Novelty was associated with subsequent portfolio returns.

The financial analysis also does not include market-wide control variables. Consequently, the estimated relationship between Coverage and subsequent excess returns may partly reflect broader financial conditions affecting both news reporting and portfolio performance. The regressions do not establish a causal effect of news attention on returns, nor do they demonstrate that the same predictive relationship would hold for other risk categories, portfolios, or periods. The limited number of Burst observations also restricts the conclusions that can be drawn about differences between ordinary and exceptional attention regimes.

These limitations suggest possible extensions of the analysis. The measurement framework could be applied to additional news sources and risk categories, while the financial tests could be repeated for different portfolios and sample periods. Further work could also investigate whether the relationship between news-based measures and market outcomes varies with the type of risk, the characteristics of the firms, or the broader financial environment. Including market-wide control variables could help assess whether Coverage contains predictive information beyond that associated with general market conditions. A separate investment application could examine whether the observed forecasting improvements lead to economically meaningful results after accounting for trading costs and portfolio risk.

The central contribution of this thesis is therefore the connection between the measurement of emerging risks in financial news and an empirical assessment of their market relevance. The four indicators provide different perspectives on the evolution of risk-related information, while the Cyber Risk application shows that one of these perspectives, the intensity of news coverage, contains measurable predictive information for weekly portfolio excess returns. The results support the usefulness of news-based risk measurement while also showing the importance of distinguishing descriptive information about risk emergence from statistically supported financial relationships.

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